
Power-Boosted Granger-Causal Discovery for Large Heterogeneous Panel Data

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Abstract

This paper proposes a power-enhanced panel Granger causality test (PE-PGCT) for assessing the Granger non-causality in heterogeneous and potentially high-dimensional panel data. Building on any existing panel Granger non-causality test, we show, both theoretically and empirically, that the proposed PE-PGCT boosts its power substantially. The power gains are particularly significant in situations of high-dimensional panels when the cross-sectional dimension exceeds the time dimension, as well as under sparse alternatives when the signals are sparsely distributed across panel units. We establish rigorous theoretical guarantees on the asymptotic behavior of the proposed power enhancement component, demonstrating attractive power enhancement properties that it induces negligible size distortion under the null hypothesis while delivering significant power gain under the alternatives. The empirical performances are illustrated via extensive simulation studies, as well as a real-world application.

1. Introduction

Granger causality, originally introduced by Granger (1969), has become a foundational tool for uncovering dynamic relationships and directional predictability in time series. It defines causality in a predictive sense (Granger, 1980; 2003; 2004): one series is said to Granger-cause another if its past values improve the forecast of the dependent variable beyond the information contained in the target’s own past. With the growth of large-scale longitudinal data, panel Granger causality test (PGCT), which extends Granger causality to panel settings by exploiting both cross-sectional and temporal dimensions, has attracted increasing attention.

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Early developments of PGCT are closely linked to the literature on dynamic panel models and panel vector autoregressions. The seminal work of Holtz-Eakin et al. (1988) assumes homogeneous dynamics across cross-sectional units, and examines Granger causality by pooling data to estimate a common autoregressive structure. Recognizing such homogeneity assumption may be too restrictive in practical applications, subsequent research (Kónya, 2006; Emirmahmutoglu & Kose, 2011) has evolved toward more flexible approaches that allow coefficients to vary across units, thereby accounting for cross-sectional heterogeneity.

One major methodological breakthrough in PGCT is the Dumitrescu-Hurlin (DH) test (Dumitrescu & Hurlin, 2012), which has become one of the most widely used methods among practitioners. The DH test estimates unit-specific autoregressive models, computes individual Wald statistics for the null hypothesis of no causality, and then averages these statistics across the panel. Under suitable regularity conditions, the standardized average converges in distribution to a normal limit under a sequential asymptotic regime when the time dimension $T \rightarrow \infty$ first, followed by the cross-sectional dimension $N \rightarrow \infty$. This asymptotic property makes the DH appealing for panels with large T , while limiting its ability to control Type I error in panels with small or moderate T , especially when $T < N$.

To address this limitation, Juodis et al. (2021) propose the Half-Panel Jackknife (HPJ) test, which challenges the necessity of averaging unit-level statistics. Their key insight is that under the null hypothesis of non-causality, the Granger-causal coefficients are identically zero for all units, rendering the parameter of interest homogeneous under the null, even when other model parameters remain heterogeneous. By exploiting this structure, HPJ employs a pooled least-squares estimator and a quadratic-form statistic, achieving a faster convergence rate and improving finite-sample performance in panels with large N . Several subsequent studies have further facilitated practical implementation (Lopez & Weber, 2017; Xiao et al., 2023) and extended the PGCT methodology in various directions (Minorics et al., 2022; Nazlioglu & Karul, 2024).

Despite their popularity, existing PGCT methods face intrinsic limitations. The DH test aggregates evidence by averaging individual Wald statistics across panel units, while

the HPJ test relies on a quadratic-form aggregation of the estimated Granger-causal coefficients. Such aggregation schemes are well-suited to situations with dense signals, but may substantially dilute strong signals when only a small subset of panel units exhibits nonzero Granger-causal effects, resulting in severe power loss under such alternatives. Consistent with the terminology in hypothesis testing literature, we refer to this type of alternative as the “**sparse alternatives**” (Fan et al., 2015; Yu et al., 2024a;b), when the number of signal-bearing coordinates is small. In modern applications involving large panels with heterogeneous dynamics, e.g., firm-level financial data or user-level behavior logs, sparse alternatives are prevalent. Hence, this limitation becomes increasingly problematic, motivating the need for methods that boost power under sparse alternatives.

In this work, we propose a new PE-PGCT, that is specifically designed for large-scale heterogeneous panels and delivers substantial power gains, particularly under sparse alternatives. Recognizing the fundamental limitation that informative signals are often obscured by cross-sectional aggregation through simple averaging or quadratic-form aggregation, our approach boosts the detection power by selectively amplifying informative marginal signals. We construct a data-driven screening component, named “PE component”, and integrate it with an existing PGCT. As a result, the resulting PE-PGCT adaptively accumulates evidence from individual units exhibiting potential Granger-causal effects for power boosting.

The proposed PE-PGCT enjoys several appealing theoretical and practical properties. We establish rigorous guarantees on its validity, showing that it achieves substantial power gains under sparse alternatives while maintaining correct Type I error control under the null. Moreover, PE-PGCT is provably at least as powerful as the baseline PGCT, ensuring no loss of power in any circumstance. Moreover, our framework is flexible and can be combined with a broad family of baseline PGCTs. On a separate note, existing PGCTs, such as DH and HPJ, exhibit markedly different performance across panel regimes; see an empirical illustration in Figure 1. These differences are rooted in the distinct asymptotic regimes under which the two tests are valid. The proposed PE-PGCT can be built on any existing PGCT and consistently boosts its power, since the PE strategy itself remains valid regardless of the choice of baseline PGCT. Additionally, we provide practical guidelines for selecting a data-adaptive baseline PGCT, which is shown to deliver appealing performance across a wide range of panel regimes.

2. Methods

2.1. Model Setup

Consider a panel with N units observed over T time periods. Let $\{x_{i,t}\}$ and $\{y_{i,t}\}$ denote the covariate and response of

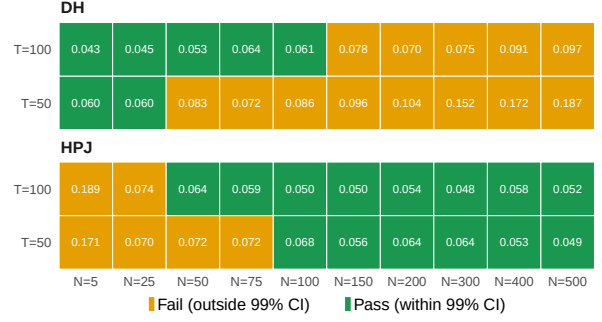


Figure 1. Failure map of Type I error control for DH and HPJ (inside vs. outside the 99% confidence intervals) based on 1,000 simulations. The empirical results shows DH is reliable for large T and small N but over-rejects when N is large, whereas HPJ is reliable for large N and small T but over-rejects when N is small.

individual i at time point t , $i = 1, \dots, N$, $t = 1, \dots, T$. Here, we allow the cross-sectional dimension N to be very large, even larger than the time dimension T . Consider a linear dynamic panel model

$$y_{i,t} = \alpha_i + \sum_{j=1}^K \gamma_{ij} y_{i,t-j} + \sum_{j=1}^K \beta_{ij} x_{i,t-j} + \epsilon_{i,t}, \quad (1)$$

where α_i is a panel-specific heterogeneous effect that is time-invariant but allowed to take different values across panel units, γ_{ij} and β_{ij} are heterogeneous coefficients that are also allowed to differ across panel units, $\epsilon_{i,t}$ is the idiosyncratic errors that are assumed to be independent with mean 0 and variance σ_i^2 . This model can accommodate cross-sectional heteroscedasticity by allowing σ_i^2 to take different values for $i = 1, \dots, N$.

K is the lag order. Throughout this work, we assume K is small and finite. Though it is not necessary that the lag orders for $\{x_{i,t}\}$ and $\{y_{i,t}\}$ are identical, we follow common practice by setting them to take equal values (Holtz-Eakin et al., 1988). Here, we assume K is known. The selection of the lag length K has been well studied in the literature, e.g., Lütkepohl (1985); Ivanov & Kilian (2005); Bruns & Stern (2019); Shojaie & Fox (2022).

Under the formulation of (1), the problem of testing Granger non-causality of $\{x_{i,t}\}$ on $\{y_{i,t}\}$ becomes testing

$$H_0 : \beta_i = (\beta_{i1}, \dots, \beta_{iK})^\top = \mathbf{0} \text{ for all } i = 1, \dots, N \\ \text{vs } H_a : \text{at least one } \beta_i \text{ is not zero} \quad (2)$$

Notation. Before proceeding, we first introduce some necessary notation. We use c or C to denote generic positive constants, which may vary from line to line. For any integer n , we use $[n]$ to denote the set $\{1, \dots, n\}$. For two sequences $\{a_n\}$ and $\{b_n\}$, we use $a_n \lesssim b_n$ to denote $a_n \leq Cb_n$ for some universal constant $C > 0$ that does not depend on n , and we use $a_n \asymp b_n$ to denote $a_n \lesssim b_n$ and $b_n \lesssim a_n$.

For any random variable $U \in \mathbb{R}$, the sub-Gaussian norm is defined as $\|U\|_{\psi_2} = \inf \{c > 0 : \mathbb{E}[\exp(U^2/c^2)] \leq 2\}$. For any random vector $\mathbf{V} \in \mathbb{R}^d$, the sub-Gaussian norm is defined as $\|\mathbf{V}\|_{\psi_2} = \sup_{\boldsymbol{\nu} \in \mathcal{S}^{d-1}} \|\boldsymbol{\nu}^\top \mathbf{V}\|_{\psi_2}$, where $\mathcal{S}^{d-1} = \{\boldsymbol{\nu} \in \mathbb{R}^d : \|\boldsymbol{\nu}\|_2 = 1\}$ is the unit sphere in $\mathbb{R}^d$.

2.2. Power Enhancement (PE)

Let $T_{pivotal}$ denote any existing pivotal panel Granger-causality test that attains correct asymptotic size. In what follows, we propose PE-PGCT, a power-enhanced panel Granger-causality test, which substantially boosts the statistical power relative to $T_{pivotal}$, while incurring asymptotically negligible impact on Type I error control.

To achieve this, we propose a power enhancement component J_{PE} , tailored to the panel Granger-causality setting. This PE component is constructed to take the value zero under H_0 to maintain proper Type I error control, but to diverge rapidly under H_a to amplify power. Though we adopt the concept of ‘‘power enhancement’’ to relate to existing literature, the proposed technique introduces new methodological aspects beyond prior works. Specifically, directly adapting the power enhancement method of Fan et al. (2015) to the panel Granger causality test presents several challenges, while our approach overcomes those limitations. A detailed comparison is provided in Appendix C.3 to more explicitly demonstrate our contribution and innovation.

We begin by extracting marginal signal information from each panel unit. We resort to the individual Wald statistic, denoted by W_i , for testing $H_{0i} : \beta_i = \mathbf{0}$, respectively, for each $i \in [N]$. The popular Dumitrescu-Hurlin (DH) test (Dumitrescu & Hurlin, 2012) aggregates these statistics by averaging, which provides a natural and stable way to pool information across multiple panel units, whereas such averaging may dilute strong but localized signals, thereby weakening the test power when Granger causality is present in only a subset of units. To more effectively detect the presence of such signals, we focus on the extreme behavior of the individual statistics and consider their maximum:

$$M_{Wald} = \max_{1 \leq i \leq N} W_i. \quad (3)$$

Equipped with M_{Wald} , we define the PE component as

$$J_{PE} = \sqrt{N} \cdot M_{Wald} \cdot I\{M_{Wald} > \delta_{N,T}\} \quad (4)$$

where $\delta_{N,T}$ is a carefully-chosen threshold to separate estimation noise from genuine signals. It plays a critical role in balancing Type-I error control and power enhancement. Intuitively, $\delta_{N,T}$ should slightly exceed the stochastic order of M_{Wald} under H_0 . The indicator function in (4) further ensures that J_{PE} is activated only when M_{Wald} exceeds the noise level implied by the null distribution.

Remark 2.1 (Strength of J_{PE} under sparse alternatives). J_{PE} is particularly effective under sparse alternatives, where Granger causality is present in only a small subset of panel units. Intuitively, M_{Wald} acts as a signal detector. By construction, M_{Wald} is highly sensitive to strong marginal signals even when signals are rare across panels. Under sparse alternatives, J_{PE} diverges at the rate $\sqrt{N}$ as long as one marginal signal is captured by M_{Wald} ; see Theorem 3.8 for formal characterization of high-power regions of J_{PE} . This makes J_{PE} an effective complement to existing panel Granger-causality tests, which primarily rely on averaging strategy and favor dense alternatives.

Although under some regularity conditions, each individual W_i converges in distribution to a χ_K^2 distribution as $T \rightarrow \infty$ under the respective null hypothesis H_{0i} , the asymptotic behavior of their maximum is considerably more delicate, especially when N diverges with T and even more so in modern settings when N may exceed T . As we show in Proposition 3.5, characterizing the asymptotic distribution of $\max_{1 \leq i \leq N} W_i$ and its deviation to $\max_{1 \leq i \leq N} \chi_{K,i}^2$ becomes technically challenging in such regimes.

After carefully examining the asymptotic distribution of M_{Wald} , we define the threshold

$$\delta_{N,T} = (K + 2\sqrt{K \log N} + 2 \log N) \cdot C(T), \quad (5)$$

where $C(T)$ is a slowly diverging correction factor given by

$$C(T) = \max\{1.5, \log(\log(T))\}, \quad (6)$$

which accounts for finite-sample deviations and ensures uniform control of the null behavior as $N, T \rightarrow \infty$. The constant 1.5 in $C(T)$ is a practical calibration choice to safeguard the finite-sample performance, aiming to provide a simple, stable, and broadly applicable implementation rule. Theoretically, any choice of $C(T) > 1$ such that $C(T) \rightarrow \infty$ as $T \rightarrow \infty$ (albeit arbitrarily slowly) would suffice for establishing the asymptotic theoretical guarantees. However, in finite samples with small or moderate T , $\log \log(T)$ takes small values, potentially leading to under-thresholding and hence distorted Type I error. The lower bound of 1.5 is therefore imposed to ensure a minimum level of thresholding that stabilizes the test performance in small and moderate samples.

In Theorem 3.8, we prove that this construction guarantees that the PE component J_{PE} is asymptotically negligible under H_0 , but diverges under alternatives whenever at least one panel unit exhibits strong Granger causality.

It is important to note that the PE component J_{PE} itself does not work as a test statistic. Rather, its role is to boost the testing power over an existing test after being added to a pivotal test $T_{pivotal}$. The asymptotic null distribution of the PE-PGCT is governed by $T_{pivotal}$, not by J_{PE} . We define

the test statistic for the proposed Power-Enhanced Panel Granger Causality Test (PE-PGCT) as

$$T_{PE-PGCT} = T_{pivotal} + J_{PE}. \quad (7)$$

Given any significance level α , let $q_{pivotal}^\alpha$ denote the critical value of the test $T_{pivotal}$, that is, $P(T_{pivotal} \geq q_{pivotal}^\alpha | H_0) \leq \alpha$. The theoretical analysis in Section 3 proves that $\mathbb{P}(J_{PE} = 0 | H_0) \rightarrow 1$. As a result, we note that the addition of J_{PE} does not affect the asymptotic null distribution, and the limiting null distribution of $T_{PE-PGCT}$ coincides with that of $T_{pivotal}$. Consequently, PE-PGCT rejects H_0 if

$$T_{PE-PGCT} \geq q_{pivotal}^\alpha. \quad (8)$$

To provide a ready-to-use tool for practitioners, we offer recommendations for the choice of the pivotal test below. This recommendation is based on a careful assessment of the relative strengths and limitations of existing approaches in the literature. As shown in Figure 1, DH performs well when $N < T - K$, but exhibits severely inflated Type I error when $N \geq T - K$, whereas HPJ shows the opposite pattern. Remark 2.2 is therefore designed to ensure that the baseline remains a valid pivotal test in each regime.

Remark 2.2 (The choice of the pivotal test $T_{pivotal}$). Methodologically, $T_{pivotal}$ can be instantiated using any existing pivotal panel Granger-causality test with correct asymptotic size. After carefully weighing the relative advantages and limitations of available approaches in the literature, we adopt a data-adaptive strategy and recommend different pivotal tests depending on the dimensionality of the panel. Specifically, we propose using

$$T_{pivotal} = \begin{cases} |T_{DH}|, & \text{if } N < T - K \\ T_{HPJ}, & \text{if } N \geq T - K \end{cases} \quad (9)$$

where the first regime corresponds to the traditional panel setting and the second corresponds to the high-dimensional panel regime. Mathematical details of $T_{pivotal}$ are in Appendix B. This data-adaptive choice leverages existing methods in the literature within their respective regimes of strength, thereby ensuring robustness and validity of the proposed procedure across a broad range of panel settings.

3. Theoretical Properties

In this section, we explore the theoretical properties of the proposed PE-PGCT. Proofs of all propositions and theorems are presented in Appendix A.

Assumption 3.1. For $i \in [N]$ and $t \in [T]$, let $\mathcal{F}_{i,t-1} := \sigma(\{(y_{i,s}, x_{i,s})_{s \leq t-1}\})$ denote the σ -algebra generated by the past history $(y_{i,s}, x_{i,s})$ up to time $t - 1$. For each unit $i \in [N]$, the individual innovations $\{\epsilon_{i,t}\}$ are independent, sub-Gaussian with $\|\epsilon_{i,t}\|_{\psi_2} \leq K_\epsilon < \infty$ and satisfies:

$\epsilon_{i,t} \perp \mathcal{F}_{i,t-1}$, $\mathbb{E}(\epsilon_{i,t}) = 0$, $\mathbb{E}(\epsilon_{i,t}^2) = \sigma_i^2$ with $c < \sigma_i^2 < \infty$ for some $c > 0$, and $\mathbb{E}(\epsilon_{i,t}^4) < \infty$.

Assumption 3.2. For each $i \in [N]$, the process $\{(y_{i,t}, x_{i,t})_{t \in \mathbb{Z}}\}$ is stationary and ergodic.

Assumption 3.3. For any $i \neq j$, $i, j \in [N]$, $\sigma(\{(y_{i,t}, x_{i,t}, \epsilon_{i,t})_{t \in \mathbb{Z}}\}) \perp \sigma(\{(y_{j,t}, x_{j,t}, \epsilon_{j,t})_{t \in \mathbb{Z}}\})$.

Assumption 3.4. Let $\mathbf{z}_{i,t} = (y_{t-1}, \dots, y_{t-K}, x_{t-1}, \dots, x_{t-K})^\top \in \mathbb{R}^{2K}$. We assume $\mathbf{z}_{i,t}$ follows a sub-Gaussian distribution with $\|\mathbf{z}_{i,t}\|_{\psi_2} \leq K_z < \infty$. Without loss of generality, we assume $\mathbb{E}(\mathbf{z}_{i,t}) = \mathbf{0}$ after demeaning with respect to the individual intercept. Additionally, $\mathbf{Q}_i = \mathbb{E}(\mathbf{z}_{i,t} \mathbf{z}_{i,t}^\top)$ exists and is positive definite. Furthermore, there exist positive constants $c_1, c_2 > 0$, such that $c_1 \leq \lambda_{\min}(\mathbf{Q}_i) \leq c_2$.

Assumption 3.1 outlines regularity conditions on the individual innovations, and Assumption 3.2 requires stationarity and ergodicity of the individual time-series processes. Stationarity is imposed at the unit level and accommodates heterogeneity across units, and ergodicity guarantees that each unit provides sufficient time-series information for consistent estimation. These conditions are standard and commonly adopted in dynamic panel models (Fernández-Val & Lee, 2013). Furthermore, the sub-Gaussian tail condition provides a relaxation to the Gaussian error assumption that is prevalent in the literature of Granger causality tests (Dumitrescu & Hurlin, 2012; Minorics et al., 2022; Shojaie & Fox, 2022). Assumption 3.3 states cross-sectional independence, which is also common in the literature of panel Granger causality tests (Emirmahmutoglu & Kose, 2011; Dumitrescu & Hurlin, 2012; Juodis et al., 2021). Assumption 3.4 concerns the regularity of the stacked lag vector $\mathbf{z}_{i,t}$, which collects past outcomes and covariates. These conditions are standard in high-dimensional regression settings and play a crucial role in the Gaussian approximation for high-dimensional maxima (Chernozhukov et al., 2013).

With these assumptions, we study the asymptotic properties of the proposed PE component J_{PE} and establish the asymptotic guarantees on the proposed PE-PGCT. It is worth mentioning that, although it appears natural that each individual Wald statistic W_i asymptotically converges in distribution to a χ_K^2 random variable, it is substantially more challenging to characterize the asymptotic distribution of $\max_{1 \leq i \leq N} W_i$ and its deviation to $\max_{1 \leq i \leq N} \chi_{K,i}^2$, particularly when N diverges. Proposition 3.5 addresses this challenge.

Proposition 3.5. With Assumptions 3.1 – 3.4, under the null hypothesis H_0 , we have

$$\begin{aligned} & \sup_{x > 0} \left| \mathbb{P} \left(\max_{1 \leq i \leq N} W_i \geq x \right) - \mathbb{P} \left(\max_{1 \leq i \leq N} \chi_{K,i}^2 \geq x \right) \right| \\ & \lesssim \frac{\log^{7/6}(NT)}{T^{1/6}} + \frac{\log^{3/2} N}{\sqrt{T}} + \frac{\log^{7/4} N}{T^{5/4}} + \frac{1}{N^2} \end{aligned}$$

where $\max_{1 \leq i \leq N} \chi_{K,i}^2$ denotes the maximum of N independent χ_K^2 random variables.

Proposition 3.5 provides a non-asymptotic upper bound on the Kolmogorov distance between $\max_{1 \leq i \leq N} W_i$ and $\max_{1 \leq i \leq N} \chi_{K,i}^2$. The term $\frac{\log^{7/6}(NT)}{T^{1/6}}$ arises from the Gaussian approximation error, $\frac{\log^{3/2} N}{\sqrt{T}}$ results from the random design linearization error, and $\frac{\log^{7/4} N}{T^{5/4}}$ reflects the studentization error. N^{-2} is a remainder term accounting for the low-probability “bad” events induced by estimators in the intermediate steps, which vanishes quickly as $N \rightarrow \infty$.

Remark 3.6 (A standalone max-form test based on M_{Wald}). Though max-form tests have been studied in panel econometrics and high-dimensional statistics, they have not yet been explored in the relatively understudied area of panel Granger-causality testing. As an add-on contribution, Proposition 3.5 can be used to derive the asymptotic distribution of the max-form statistic $M_{Wald} := \max_{1 \leq i \leq N} W_i$ for testing panel Granger-causality, which may itself serve as a standalone testing procedure. We present detailed discussions in Appendix C.3.

Proposition 3.7. *With Assumptions 3.1 – 3.4, under the null hypothesis H_0 , we have*

$$\begin{aligned} & \mathbb{P}(\max_{1 \leq i \leq N} W_i \geq \delta_{N,T}) \\ & \lesssim \frac{1}{\sqrt{N}} + \frac{\log^{7/6}(NT)}{T^{1/6}} + \frac{\log^{3/2} N}{\sqrt{T}} + \frac{\log^{7/4} N}{T^{5/4}} \end{aligned}$$

Proposition 3.7 provides a non-asymptotic upper bound on the probability that J_{PE} selects spurious signals under the null hypothesis. This proposition forms the theoretical foundation of PE-PGCT in ensuring the proper Type-I error rate control after adding J_{PE} to the pivotal test.

Theorem 3.8 (Asymptotic Properties of the PE-component J_{PE}). *With Assumptions 3.1 – 3.4, and suppose $\log N = o(T^{1/7})$, J_{PE} satisfies the following properties:*

- (i) *Non-negativity:* $J_{PE} \geq 0$.
- (ii) *No size distortion:* $\mathbb{P}_{H_0}(J_{PE} = 0) \rightarrow 1$ as $N, T \rightarrow \infty$, where $\mathbb{P}_{H_0}(\cdot) = \mathbb{P}(\cdot | H_0)$ denotes the probability under the null hypothesis H_0 .
- (iii) *Fast divergence under certain alternatives:* consider the alternative space Θ_β defined as

$$\begin{aligned} \Theta_\beta &= \{(\beta_1, \dots, \beta_N) : \\ & \max_{1 \leq i \leq N} \beta_i^\top \Sigma_i^{-1} \beta_i > \frac{\eta \cdot \log \log T \log N}{T} \\ & \text{for some } \eta > 2\}, \end{aligned} \quad (10)$$

for any $\beta \in \Theta_\beta$, we have $J_{PE} \rightarrow \infty$ as $N, T \rightarrow \infty$.

Theorem 3.8 presents the asymptotic properties of the proposed PE component. The three theoretical properties of J_{PE} are essential for guaranteeing the power enhancement

properties of the proposed PE-PGCT, formalized in Theorem 3.10. Importantly, the alternative space Θ_β characterizes a sufficient condition for consistent detection of J_{PE} : it shows that the presence of a single panel unit with signal strength exceeding the $(\eta \log \log T \log N)/T$ threshold is enough to trigger the divergence of J_{PE} . Consequently, Θ_β defines a class of sparse alternatives under which PE-PGCT is guaranteed to achieve asymptotically consistent power.

Theorem 3.9 (Asymptotic Null Distribution of the PE-PGCT Statistic). *With Assumptions 3.1 – 3.4, and suppose $\log N = o(T^{1/7})$, the addition of the PE component J_{PE} does not affect the limiting null distribution of the pivotal test $T_{pivotal}$, i.e., under the null hypothesis,*

$$T_{PE-PGCT} \stackrel{d}{=} T_{pivotal}$$

Theorem 3.9 characterizes the asymptotic null distribution of the PE-PGCT statistic. This theorem is a natural result of Property (ii) of Theorem 3.8. It reflects that $T_{PE-PGCT}$ shares the same limiting null distribution as $T_{pivotal}$, thereby confirming that adding J_{PE} does not distort the Type-I error control. In the subsequent theorem, we demonstrate that adding J_{PE} can substantially boost the testing power under the designated alternative space.

Theorem 3.10 (Power Enhancement Properties of PE-PGCT). *Suppose $\Theta := \{(\beta_1, \dots, \beta_N) : \beta_i \in \mathbb{R}^K, \forall i = 1, \dots, N\}$ denotes the parameter space, and $\Theta_0 = \{(\beta_1, \dots, \beta_N) : \beta_i = \mathbf{0}, \forall i = 1, \dots, N\}$ denotes the null parameter space. Let $\phi_{PE-PGCT}(\beta) = \mathbb{P}_\beta(T_{PE-PGCT} \geq q_{pivotal}^\alpha)$ denote the power function of PE-PGCT, and let $\phi_{pivotal}(\beta) = \mathbb{P}_\beta(T_{pivotal} \geq q_{pivotal}^\alpha)$ denote the power function of the pivotal test $T_{pivotal}$. The following power enhancement properties hold:*

- (i) *Type-I error rate control:* for any $\beta \in \Theta_0$, $\phi_{PE-PGCT}(\beta) \rightarrow \alpha$ as $N, T \rightarrow \infty$;
- (ii) *Monotonicity (No power loss):* for any $\beta \in \Theta$, $\phi_{PE-PGCT}(\beta) \geq \phi_{pivotal}(\beta)$;
- (iii) *Boosted power:* $\forall \beta \in \Theta_\beta$ where Θ_β is the sparse alternative space defined in (10), $\phi_{PE-PGCT}(\beta) \rightarrow 1$ as $N, T \rightarrow \infty$.

Theorem 3.10 establishes the PE properties of the proposed PE-PGCT. Property (i) guarantees no size distortion under the null. Property (ii) ensures no power loss, i.e., PE-PGCT is at least as powerful as the pivotal test. Property (iii) implies that PE-PGCT achieves consistent power as long as one marginal signal exceeds the threshold elaborated in Θ_β . Furthermore, we note that the three PE properties are fully aligned with the general PE principles outlined by Fan et al. (2015), the seminal work in the PE literature.

Table 1. Empirical Type I error under the Gaussian error setting (nominal level $\alpha = 0.05$).

T	N	PE-PGCT	DH	HPJ	T	N	PE-PGCT	DH	HPJ	T	N	PE-PGCT	DH	HPJ
50	5	0.060	0.060	0.171	100	5	0.044	0.043	0.189	200	5	0.061	0.061	0.139
	10	0.052	0.052	0.104		10	0.041	0.041	0.105		10	0.052	0.052	0.099
	20	0.062	0.062	0.087		20	0.056	0.056	0.101		20	0.060	0.059	0.084
	50	0.074	0.083	0.072		50	0.053	0.053	0.064		50	0.065	0.065	0.067
	100	0.072	0.086	0.068		100	0.050	0.061	0.050		100	0.059	0.059	0.052
	200	0.069	0.104	0.064		200	0.056	0.070	0.054		200	0.057	0.058	0.057
500	500	0.052	0.187	0.049	500	500	0.052	0.097	0.052	500	500	0.053	0.054	0.053

4. Simulation Studies

In this section, we carry out simulations to evaluate the finite-sample performance of PE-PGCT. We adopt the same data-generating process as in [Dumitrescu & Hurlin \(2012\)](#) and begin with $K = 1$. For $i = 1, \dots, N$, we generate $\{x_{i,t}\}$ from an AR(1) process with the autoregressive coefficient ϕ_i , and subsequently, $\{y_{i,t}\}$ following $y_{i,t} = \alpha_i + \gamma_i y_{i,t-1} + \beta_i x_{i,t-1} + \epsilon_{i,t}$, where the individual-specific parameters $\alpha_i, \phi_i, \gamma_i$ are generated from $\mathcal{U}[0.2, 0.8]$. We generate the innovations $\{\epsilon_{i,t}\}$ from different distributions, including a Gaussian error setting $\epsilon_{i,t} \sim \mathcal{N}(0, \sigma_{\epsilon,i}^2)$ with each $\sigma_{\epsilon,i}^2$ randomly generated from $\mathcal{U}(0.5, 1.5)$ to mimic cross-sectional heteroscedasticity, as well as a non-Gaussian setting $\epsilon_{i,t} \sim \text{standardized } t_3$. Due to space limit, non-Gaussian cases are reported in Appendix C.1. We compare PE-PGCT¹ with the Dumitrescu-Hurlin (DH) test ([Dumitrescu & Hurlin, 2012](#)) and Half-Panel Jackknife (HPJ) test ([Juodis et al., 2021](#)). A comparison of computation time is provided in Appendix C.2. Additionally, we compare our approach with the max-form test in Remark 3.6 and the method of [Fan et al. \(2015\)](#); the details are presented in Appendix C.3. All tests are conducted at the nominal level $\alpha = 0.05$. For each experimental design, we report the empirical rejection rate based on 1,000 replications, which corresponds to the empirical Type I error under the null and empirical power under the alternatives.

Evaluation of Type I error control. We set $\beta_i = 0$ for all $i \in [N]$ to simulate the null hypothesis of no Granger causality. We consider a range of panel dimensions with $T = 50, 100, 200$ and $N = 5, 10, 20, 50, 100, 200, 500$. The empirical rejection rates are summarized in Table 1. The difference in empirical Type-I error rates between PE-PGCT and DH (for $N < T - K$) or HPJ (for $N \geq T - K$) is an equivalent measurement of the null triggering rate $\hat{\mathbb{P}}(J_{PE} > 0)$. As shown in the table, DH exhibits substantial inflated Type I errors when T is small, with the size distortion becoming more severe in panels with $T \ll N$. In contrast, HPJ suffers from noticeable size distortion when N is small. The proposed PE-PGCT, however, is able to

maintain the correct Type I error control across all settings considered.

Evaluation of Power. We consider heterogeneous alternatives by setting $\beta_i = \beta \neq 0$ for $i = 1, \dots, N_0$ and $\beta_i = 0$, for $i = N_0 + 1, \dots, N$, where N_0 represents the number of cross-sectional units with non-zero Granger-causal signals. By varying β and N_0 , we assess power performance along two directions: signal strength and signal sparsity. Since Table 1 shows DH and HPJ suffer from (sometimes severe) size distortions under certain configurations, we report size-corrected power for these two to ensure fair comparisons.

We first examine power under sparse alternatives with the number of signal-bearing coordinates N_0 set to be 1. The signal strength is varied over $\beta \in \{0.1, 0.2, \dots, 0.9\}$. We consider the same range of (N, T) configurations as in the size experiments. Figure 2 plots a representative subset of results, while the complete results are reported in Table S.2 of Appendix C. HPJ is nearly powerless under sparse alternatives. This aligns with the hypothesis testing literature that quadratic-form tests lose power under sparse alternatives ([Yu et al., 2023; 2025; Li et al., 2025; Zhang et al., 2025](#)). Evidently, PE-PGCT delivers substantial power gains. DH performs reasonably well when $T > N$, but its power deteriorates markedly when $T \leq N$. In all cases, PE-PGCT exhibits clear improvements over DH. Overall, PE-PGCT dominates the competing methods in sparse alternatives across all settings considered. These empirical findings resonate with theoretical properties established in Section 3.

Next, we evaluate power under various signal sparsity levels. We vary N_0 from 0 to 9, while fixing the individual signal strength at $\beta = 0.5$ for $T = 50$, 0.4 for $T = 100$, and 0.3 for $T = 200$. A representative subset of results is visualized in Figure 3, with complete results in Table S.3 of Appendix C. Once again, PE-PGCT consistently outperforms HPJ across all configurations considered. Relative to DH, PE-PGCT exhibits clear advantages when N_0 is small, corresponding to sparse alternatives, but slightly underperforms (or remains largely comparable) when the alternative becomes dense. This pattern is consistent with our earlier discussion: averaging-based tests such as DH are well-suited for dense alternatives but tend to suffer from

¹The implementation code is available at <https://github.com/yihenggu/PE-PGCT>.

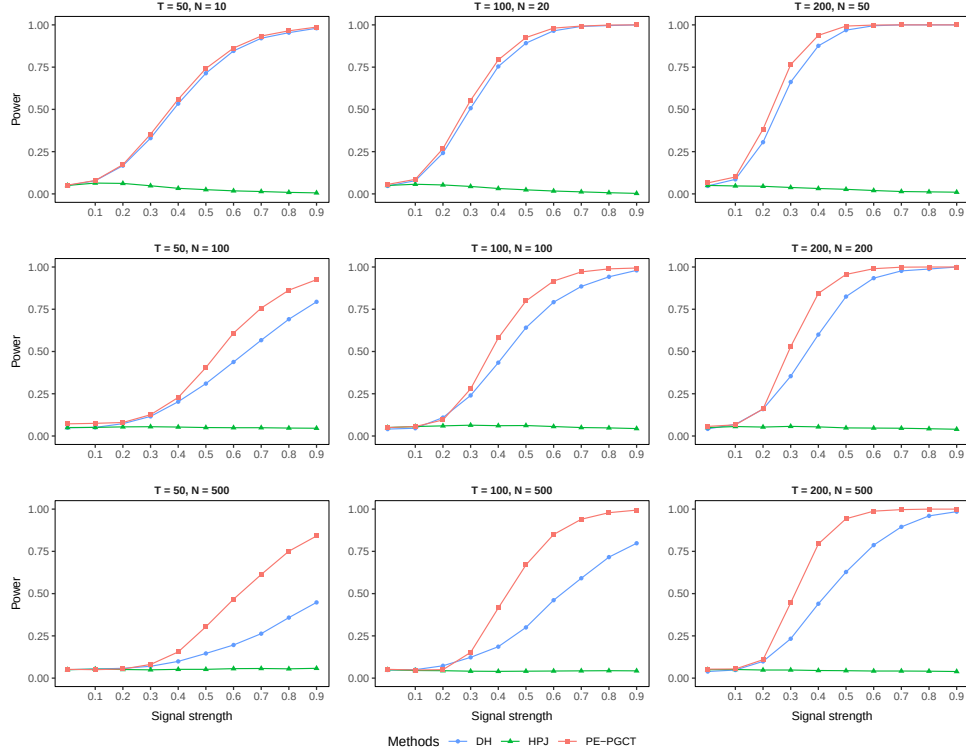


Figure 2. Empirical power under sparse alternatives ($N_0 = 1$) under the Gaussian error setting. Tests are conducted at level $\alpha = 0.05$.

power loss under sparse alternatives. It is also worth emphasizing that the DH results reported here are size-corrected power based on simulated null distributions. In practice, such size correction may not be feasible, and the asymptotic DH test is known to exhibit inflated Type I error when $T \leq N$ (as shown in Table 1). In contrast, PE-PGCT relies on an asymptotic distribution that does not require size correction, making it both convenient and reliable in empirical applications, an additional practical advantage.

5. Real Data Analysis

In this section, we investigate the Granger causal relationship between cost efficiency ($x_{i,t}$) and bank profitability ($y_{i,t}$). We use supervisory bank balance-sheet data from the Federal Deposit Insurance Corporation (FDIC) website. This dataset is publicly available and has also been used in related studies (e.g., Juodis et al. (2021)), facilitating reproducibility and comparisons across methods.

Data Preparation. $y_{i,t}$ is the profitability of bank i at time t , proxied by the return on assets (ROA). ROA is calculated by the annualized net income after taxes and extraordinary items as a percent of the average total assets. $x_{i,t}$ is the operation cost efficiency of bank i at time t . Following the standard practice of banking efficiency studies, we obtain

$x_{i,t}$ using a translog cost frontier model (Altunbas et al., 2007). Specifically, we consider

$$\begin{aligned} \log TC_{i,t} = & \eta_i + \tau_t + \nu_{i,t} + \sum_{m=1}^2 \sum_{n=1}^3 \xi_{mn} \log Q_{m,i,t} \log P_{n,i,t} \\ & + \sum_{h=1}^3 a_h \log P_{h,i,t} + \frac{1}{2} \sum_{m=1}^3 \sum_{n=1}^3 \delta_{mn} \log P_{m,i,t} \log P_{n,i,t} \\ & + \sum_{h=1}^2 b_h \log Q_{h,i,t} + \frac{1}{2} \sum_{m=1}^2 \sum_{n=1}^2 \gamma_{mn} \log Q_{m,i,t} \log Q_{n,i,t} \end{aligned}$$

where $TC_{i,t}$ refers to the total cost. $P_{1,i,t}$, $P_{2,i,t}$ and $P_{3,i,t}$ denote the capital price, labor price, and the price of loanable funds, respectively. $Q_{1,i,t}$ and $Q_{2,i,t}$ stand for the net loans and securities. The bank-specific, time-varying inefficiency is captured by $\eta_i + \tau_t$, which can be estimated using a two-way fixed effects regression. Then, the cost efficiency $x_{i,t}$ is obtained via $x_{i,t} = \exp\{\min\{\hat{\eta}_i + \hat{\tau}_t\}_{i,t} - (\hat{\eta}_i + \hat{\tau}_t)\}$.

Panel Granger Causality Tests. We analyze a panel consisting of $N = 300$ U.S. banking institutions observed quarterly from 2006:Q1 to 2019:Q4. Each institution is uniquely identified by its FDIC Bank Certificate Number (CERT). We record the CERTs and bank names in Appendix D. To assess panel Granger causality from a comprehensive perspective, we consider not only the full panel across all

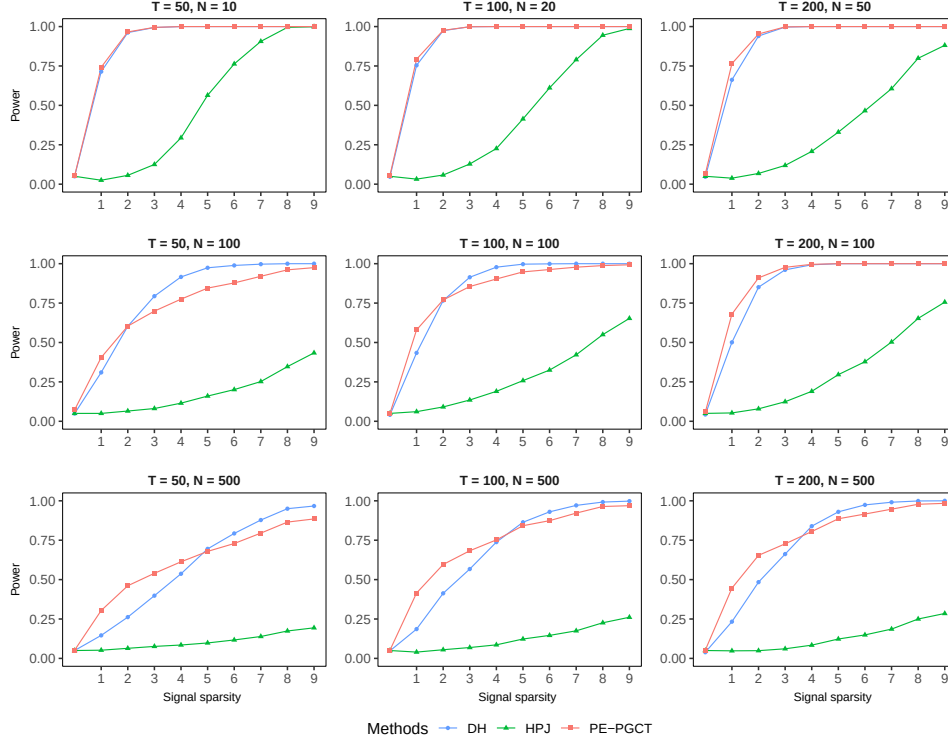


Figure 3. Empirical power in situations of different sparsity levels (signal number ranging from 0 to 9) under the Gaussian error setting. Tests are conducted at the nominal level $\alpha = 0.05$.

institutions and time periods, but also sub-panels defined along the time-series and cross-sectional dimensions.

Across the time dimension, we divide the panel into two sub-periods: 2006:Q1 - 2010:Q4 (named “Basel II”) and 2011:Q1 - 2019:Q4 (named “DFA”). This partition is economically motivated by major changes in the regulatory regime before and after the global financial crisis. The Basel II sub-period corresponds to the regulatory environment dominated by the Basel II framework, in which capital requirements relied heavily on risk-weighted assets and internal models, increasing banks’ flexibility but also their exposure to model risk and procyclicality. This period also includes the 2007–2009 global financial crisis, which exposed major weaknesses of Basel II and represents pre- and crisis-era banking behavior before comprehensive post-crisis reforms were implemented. The DFA sub-period reflects the post-crisis regulatory regime shaped primarily by the Dodd–Frank Act, which became effective from 2011 onward. This period is characterized by tighter supervision, stress testing, higher capital and liquidity standards, and enhanced oversight of systemically important banks. These reforms fundamentally changed banks’ incentives and risk-taking behavior, making this sub-period a natural post-reform comparison window.

Along the cross-sectional dimension, we divide banks into two groups using the k -means algorithm based on their average bank size, proxied by the log total assets. This procedure yields 220 small banks and 80 large banks. Summary statistics for the two groups are reported in Table S.9.

To assess the Granger causality, we set the lag order to be 1 and employ the pivotal test defined in (9). We compare PE-PGCT with the pivotal test across the full panel and sub-panels defined above. The p -values are reported in Table 2, along with the banks identified by the PE component J_{PE} .

Empirical Findings. Table 2 shows PE-PGCT and the pivotal test reach consistent conclusions across all settings. We also observe that PE-PGCT is at least as powerful as the pivotal test, which is as expected by design.

The results suggest that cost efficiency Granger-causes profitability at the significance level of 1% in nearly all cases, with the only exception being that large banks temporarily decoupled profits from efficiency under Basel II but this link is restored under tighter post-crisis DFA regulation. These findings are well aligned with the mainstream empirical literature. A large body of literature has shown that operational efficiency is a primary driver of bank profitability (Berger & Mester, 1997; Berger & Humphrey, 1997; Athanasoglou et al., 2008). The absence of causality for large banks dur-

Table 2. p -values of the PE-PGCT and Pivotal tests, along with the PE component J_{PE} and banks identified by J_{PE} .

Group	Period	J_{PE}	PE-PGCT	Pivotal	Banks identified by J_{PE}
All	Full	489.313	$< 10^{-16}$	1.1×10^{-16}	Elevate Bank (CERT 4770), Hilltop National Bank (CERT 19184)
	Basel II	0	$< 10^{-16}$	$< 10^{-16}$	–
	DFA	452.419	$< 10^{-16}$	$< 10^{-16}$	First State Bank of San Diego (CERT 17350)
Small	Full	419.023	$< 10^{-16}$	1.1×10^{-12}	First National Bank of Bellville (CERT 3099), Mid Penn Bank (CERT 9889)
	Basel II	0	$< 10^{-16}$	$< 10^{-16}$	–
	DFA	387.428	$< 10^{-16}$	4.5×10^{-12}	Solvay Bank (CERT 13297)
Large	Full	248.584	$< 10^{-16}$	9.3×10^{-10}	First National Bank of Milaca (CERT 5198)
	Basel II	0	0.011	0.011	–
	DFA	0	1.1×10^{-4}	1.1×10^{-4}	–

ing the Basel II period aligns with evidence that large-bank profitability before the financial crisis was driven more by risk-taking, trading income, and funding advantages associated with size than by operating efficiency (DeYoung & Torna, 2013; Juodis et al., 2021; Cui et al., 2023). The re-emergence of the causal relationship in the DFA period is consistent with post-crisis regulatory reforms that reduced the “Too Big to Fail” advantage and restored the importance of managerial efficiency for profitability (Wheelock & Wilson, 2018; Cetorelli & Traina, 2021).

Moreover, by incorporating the PE component, PE-PGCT uncovers additional findings of Granger causality at the unit level, offering finer-grained insights and revealing new aspects of the underlying dynamics. For example, during the DFA period, First State Bank of San Diego (FSBSD, CERT 17350) is identified from our test among all banks. The findings support the “Bad Management Hypothesis”, illustrated in the work of Berger & DeYoung (1997), that low cost efficiency Granger-causes an increase of non-performing loans, which subsequently affects the profit. As a small community bank, when the influence of significant compliance costs increased due to the DFA, marginal changes in operating expenses reacted more quickly and were a more direct and critical predictor of earning conditions. Moreover, San Diego’s rising unemployment in late 2011 impacted banking and business, forcing FSBSD to manage cost controls. Meanwhile, when facing an increasing burden of post DFA era, Solvay Bank (CERT 13297) is identified among small banks during 2011-2019. Similarly, as a lean community bank, Solvay Bank is strongly focused on local lending, which enhances the predictive impact of operational control on profitability. For the full period, the Elevate Bank (previously named as First National Bank of Sedan before 2023, CERT 4770), the First National Bank of Bellville (CERT 3099), and the First National Bank of Milaca (CERT 5198) are identified among all banks, small banks, and large banks, respectively. The possible reason behind the strong signals is that this period includes the Great Recession and the post-Dodd-Frank regulatory surge, where these banks use efficiency gains or specialized lending focus to maintain profitability in spite of rising external costs.

6. Discussion

In this paper, we propose PE-PGCT, a powerful framework for assessing the Granger non-causality in large-scale heterogeneous panels that delivers substantial power gains, particularly under sparse alternatives. While we focus on the relatively underexplored problem of panel Granger causality testing to present the methodology in a focused and coherent manner, the proposed power enhancement technique is not restricted to PGCT. More broadly, it can be naturally adapted and applicable to a wide range of high-dimensional inference problems.

There are a few promising directions for future work. Consistent with related studies such as Emirmahmutoglu & Kose (2011); Dumitrescu & Hurlin (2012) and Juodis et al. (2021), we assume independent error terms to facilitate the theoretical analysis and obtain tractable asymptotic results. We acknowledge that this assumption may be somewhat restrictive in practical applications. To overcome this limitation, Emirmahmutoglu & Kose (2011); Dumitrescu & Hurlin (2012) and Xiao et al. (2023) discussed practical strategies using block bootstrap, while Juodis et al. (2021) discussed the possibility of modeling the cross-sectional dependence through latent factor structures. Though not reported in the paper, we conducted additional experiments under moderately correlated errors, and the test performance remains empirically robust. Nevertheless, developing a theoretical framework that rigorously accommodates such cross-sectional dependence is an important and nontrivial extension. It would require substantially different technical tools, and we shall leave it for future research.

Moreover, our current framework assumes stationarity, which is standard in panel Granger causality settings to ensure valid inference. In practice, non-stationarity can be potentially addressed by data preprocessing. Extending the methodology to non-stationary scenarios is an important direction for future research. In principle, we expect such an extension would require adapting the test to account for common stochastic trends or employing transformations to restore stationarity, yet a thorough investigation is beyond the scope of this work and shall be left for future work.

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Impact Statement

This paper advances methods for Granger causality analysis. Improved methods may have positive impacts across many domains, including economics, climate science, neuroscience, epidemiology, and other scientific fields where understanding temporal relationships is critical for decision making and hypothesis generation.

At the same time, Granger-causal analysis may be misapplied or over-interpreted, particularly when assumptions are violated. Incorrect Granger-causal conclusions could lead to flawed scientific inferences or poor downstream decisions in high-stakes settings such as policy making, healthcare, or finance. We therefore emphasize that the proposed method is intended as a statistical tool to support, rather than replace, domain expertise and careful experimental design.

We do not foresee direct negative societal impacts arising from this work. As with other advances in Granger-causal analysis and machine learning, responsible use requires transparency about assumptions, limitations, and uncertainty when applying these methods in real-world contexts.

References

- Altunbas, Y., Carbo, S., Gardener, E. P., and Molyneux, P. Examining the relationships between capital, risk and efficiency in european banking. *European Financial Management*, 13(1):49–70, 2007.
- Athanasoglou, P. P., Brissimis, S. N., and Delis, M. D. Bank-specific, industry-specific and macroeconomic determinants of bank profitability. *Journal of International Financial Markets, Institutions and Money*, 18(2):121–136, 2008.
- Berger, A. N. and DeYoung, R. Problem loans and cost efficiency in commercial banks. *Journal of Banking & Finance*, 21(6):849–870, 1997.
- Berger, A. N. and Humphrey, D. B. Efficiency of financial institutions: International survey and directions for future research. *European Journal of Operational Research*, 98(2):175–212, 1997.
- Berger, A. N. and Mester, L. J. Inside the black box: What explains differences in the efficiencies of financial institutions? *Journal of Banking & Finance*, 21(7): 895–947, 1997.
- Bruns, S. B. and Stern, D. I. Lag length selection and p-hacking in granger causality testing: prevalence and performance of meta-regression models. *Empirical Economics*, 56(3):797–830, 2019.
- Cetorelli, N. and Traina, J. Resolving “too big to fail”. *Journal of Financial Services Research*, 60(1):1–23, 2021.
- Chernozhukov, V., Chetverikov, D., and Kato, K. Gaussian approximations and multiplier bootstrap for maxima of sums of high-dimensional random vectors. *The Annals of Statistics*, 41(6):2786–2819, 2013.
- Chernozhukov, V., Chetverikov, D., and Kato, K. Comparison and anti-concentration bounds for maxima of gaussian random vectors. *Probability Theory and Related Fields*, 162(1):47–70, 2015.
- Chernozhukov, V., Chetverikov, D., and Kato, K. Central limit theorems and bootstrap in high dimensions. *The Annals of Probability*, 45(4):2309 – 2352, 2017.
- Cui, G., Sarafidis, V., and Yamagata, T. Iv estimation of spatial dynamic panels with interactive effects: large sample theory and an application on bank attitude towards risk. *The Econometrics Journal*, 26(2):124–146, 2023.
- DeYoung, R. and Torna, G. Nontraditional banking activities and bank failures during the financial crisis. *Journal of Financial Intermediation*, 22(3):397–421, 2013.
- Dhaene, G. and Jochmans, K. Split-panel jackknife estimation of fixed-effect models. *The Review of Economic Studies*, 82(3):991–1030, 2015.
- Dumitrescu, E.-I. and Hurlin, C. Testing for Granger non-causality in heterogeneous panels. *Economic Modelling*, 29(4):1450–1460, 2012.
- Emirmahmutoglu, F. and Kose, N. Testing for Granger causality in heterogeneous mixed panels. *Economic Modelling*, 28(3):870–876, 2011.
- Fan, J., Liao, Y., and Yao, J. Power enhancement in high-dimensional cross-sectional tests. *Econometrica*, 83(4):1497–1541, 2015.
- Fernández-Val, I. and Lee, J. Panel data models with nonadditive unobserved heterogeneity: Estimation and inference. *Quantitative Economics*, 4(3):453–481, 2013. doi: <https://doi.org/10.3982/QE75>.
- Granger, C. W. Investigating causal relations by econometric models and cross-spectral methods. *Econometrica*, 37(3): 424–438, 1969.

- Granger, C. W. Testing for causality: a personal viewpoint. *Journal of Economic Dynamics and Control*, 2:329–352, 1980.
- Granger, C. W. Some aspects of causal relationships. *Journal of Econometrics*, 112(1):69–69, 2003.
- Granger, C. W. Time series analysis, cointegration, and applications. *American Economic Review*, 94(3): 421–425, 2004.
- Holtz-Eakin, D., Newey, W., and Rosen, H. S. Estimating vector autoregressions with panel data. *Econometrica*, 56(6):1371–1395, 1988.
- Ivanov, V. and Kilian, L. A practitioner’s guide to lag order selection for var impulse response analysis. *Studies in Nonlinear Dynamics & Econometrics*, 9(1), 2005.
- Juodis, A., Karavias, Y., and Sarafidis, V. A homogeneous approach to testing for Granger non-causality in heterogeneous panels. *Empirical Economics*, 60(1): 93–112, 2021.
- Kónya, L. Exports and growth: Granger causality analysis on OECD countries with a panel data approach. *Economic Modelling*, 23(6):978–992, 2006.
- Laurent, B. and Massart, P. Adaptive estimation of a quadratic functional by model selection. *The Annals of Statistics*, 28(5):1302–1338, 2000.
- Li, D., Xue, L., Yang, H., and Yu, X. Power-enhanced two-sample mean tests for high-dimensional microbiome compositional data. *Biometrics*, 81(2):ujaf034, 2025.
- Lopez, L. and Weber, S. Testing for Granger causality in panel data. *The Stata Journal*, 17(4):972–984, 2017.
- Lütkepohl, H. Comparison of criteria for estimating the order of a vector autoregressive process. *Journal of Time Series Analysis*, 6(1):35–52, 1985.
- Minorics, L., Turkmen, C., Kernert, D., Bloebaum, P., Callot, L., and Janzing, D. Testing Granger non-causality in panels with cross-sectional dependencies. In *International Conference on Artificial Intelligence and Statistics*, pp. 10534–10554. PMLR, 2022.
- Nazlioglu, S. and Karul, C. Testing for Granger causality in heterogeneous panels with cross-sectional dependence. *Empirical Economics*, 67(4):1541–1579, 2024.
- Rudelson, M. and Vershynin, R. Hanson-Wright inequality and sub-Gaussian concentration. *Electronic Communications in Probability*, 18(82):1 – 9, 2013.
- Shojaie, A. and Fox, E. B. Granger causality: A review and recent advances. *Annual Review of Statistics and Its Application*, 9(1):289–319, 2022.
- Wheelock, D. C. and Wilson, P. W. The evolution of scale economies in us banking. *Journal of Applied Econometrics*, 33(1):16–28, 2018.
- Xiao, J., Karavias, Y., Juodis, A., Sarafidis, V., and Ditzen, J. Improved tests for granger noncausality in panel data. *The Stata Journal*, 23(1):230–242, 2023.
- Yu, X., Li, D., Xue, L., and Li, R. Power-enhanced simultaneous test of high-dimensional mean vectors and covariance matrices with application to gene-set testing. *Journal of the American Statistical Association*, 118(544): 2548–2561, 2023.
- Yu, X., Li, D., and Xue, L. Fisher’s combined probability test for high-dimensional covariance matrices. *Journal of the American Statistical Association*, 119(545):511–524, 2024a.
- Yu, X., Yao, J., and Xue, L. Power enhancement for testing multi-factor asset pricing models via Fisher’s method. *Journal of Econometrics*, 239(2):105458, 2024b.
- Yu, X., Zhang, L., Srinivasan, A., Xie, M.-G., and Xue, L. A unified combination framework for dependent tests with applications to microbiome association studies. *Biometrics*, 81(1):ujaf001, 2025.
- Zhang, Z., Yu, X., and Li, R. A novel approach of high dimensional linear hypothesis testing problem. *Journal of the American Statistical Association*, 120(552): 2235–2249, 2025.

Appendices

The appendices of this paper are organized into four parts. Appendix A contains technical lemmas and the proofs of all propositions and theorems presented in Section 3. Appendix B provides additional details on the pivotal test T_{pivotal} , supplementing the discussion in Section 2.2. Appendix C presents further simulation results that complement those in Section 4. Finally, Appendix D provides supplementary information on the data used in the real data analysis in Section 5.

A. Technical Lemmas and Proofs

A.1. Lemmas

We first present some helpful lemmas and their proofs, which help facilitate the proofs of the propositions and theorems in the main context.

Lemma A.1. *Let $M_{\chi^2} := \max_{1 \leq i \leq N} \chi_{K,i}^2$ denote the max of $N (\geq 2)$ independent χ_K^2 random variables. For any $\varepsilon > 0$, there exists $C > 0$, such that*

$$\sup_x \mathbb{P}(|M_{\chi_K^2} - x| \leq \varepsilon) \leq C\varepsilon. \quad (\text{S.1})$$

Proof of Lemma A.1. Let $U_1, \dots, U_N \stackrel{i.i.d.}{\sim} \chi_K^2$ with the cumulative density function (CDF) $F(x)$ and probability density function (PDF) $f(x)$. For convenience, we define the survival function $\bar{F}(x) = 1 - F(x)$ and the hazard function $h(x) = f(x)/\bar{F}(x)$. Then the CDF and PDF of $M_{\chi^2} := \max_{1 \leq i \leq N} U_i$ can be denoted using

$$F_{M_{\chi^2}}(x) = \mathbb{P}(M_{\chi^2} \leq x) = (F(x))^N, \quad f_{M_{\chi^2}}(x) = Nf(x)(F(x))^{N-1}.$$

For any $\varepsilon > 0$,

$$\mathbb{P}(x < M_{\chi^2} \leq x + \varepsilon) = \int_x^{x+\varepsilon} f_{M_{\chi^2}}(t) dt \leq \varepsilon \cdot \sup_t f_{M_{\chi^2}}(t).$$

Hence,

$$\sup_x \mathbb{P}(|M_{\chi_K^2} - x| \leq \varepsilon) \leq 2\varepsilon \cdot \sup_t f_{M_{\chi^2}}(t) \quad (\text{S.2})$$

It suffices to show, there exists some positive constant C , such that

$$\sup_t f_{M_{\chi^2}}(t) \leq C < \infty. \quad (\text{S.3})$$

We begin with the case $K = 1$. The CDF and PDF of χ_1^2 can be rewritten as

$$F(x) = 2\Phi(\sqrt{x}) - 1, \quad f(x) = F'(x) = \frac{\phi(\sqrt{x})}{\sqrt{x}},$$

where $\Phi(x)$ and $\phi(x) = \frac{1}{\sqrt{2\pi}} \exp(-\frac{1}{2}x^2)$ are CDF and PDF of the standard normal distribution. Rewriting $f_{M_{\chi^2}}(x)$ into the hazard form,

$$f_{M_{\chi^2}}(x) = Nf(x)(F(x))^{N-1} = h(x) \cdot N\bar{F}(x)(1 - \bar{F}(x))^{N-1}.$$

Let $g(u) := Nu(1-u)^{N-1}$, $u \in [0, 1]$. We can show that for any $u \in [0, 1]$, $\sup_{u \in [0, 1]} g(u) = g(N^{-1}) = (1 - N^{-1})^{N-1} \leq 2e^{-1}$. Therefore,

$$\sup_x f_{M_{\chi^2}}(x) \leq 2e^{-1} \sup_x h(x). \quad (\text{S.4})$$

For χ^2 distribution with degree of freedom $K = 1$, the hazard function takes the form

$$h(x) = \frac{f(x)}{\bar{F}(x)} = \frac{\phi(\sqrt{x})/\sqrt{x}}{2 - 2\Phi(\sqrt{x})} = \frac{1}{2\sqrt{x}} \cdot \frac{\phi(\sqrt{x})}{1 - \Phi(\sqrt{x})}$$

With the Mills-ratio inequality,

$$h(x) \leq \frac{1}{2\sqrt{x}} \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right) = \frac{1}{2} + \frac{1}{2x}.$$

Given this, we will bound $f_{M_{\chi^2}}(x)$ by splitting the domain into $x \in (0, 1]$ and $x \geq 1$.

For any $x \geq 1$, $h(x) \leq \frac{1}{2} + \frac{1}{2} = 1$, and hence,

$$\sup_{t \geq 1} f_{M_{\chi^2}}(t) \leq 2e^{-1} \sup_{t \geq 1} h(t) \leq 2e^{-1}. \quad (\text{S.5})$$

For any $x \in (0, 1]$, since $\phi(x) \geq \phi(0) = (2\pi)^{-\frac{1}{2}}$, we have

$$\Phi(x) = \Phi(0) + \int_0^x \phi(t) dt \leq \Phi(0) + \int_0^x \phi(0) dt = \frac{1}{2} + x \cdot \phi(0) = \frac{1}{2} + \frac{x}{\sqrt{2\pi}}$$

and naturally,

$$F(x) = 2\Phi(\sqrt{x}) - 1 \leq 2 \cdot \left(\frac{1}{2} + \frac{x}{\sqrt{2\pi}} \right) - 1 = \frac{\sqrt{2x}}{\sqrt{\pi}}$$

and

$$f(x) = \frac{\phi(x)}{\sqrt{x}} \leq \frac{\phi(0)}{\sqrt{x}} = \frac{1}{\sqrt{2\pi x}}.$$

Thus, when $x \in (0, 1]$ and $N \geq 2$,

$$f_{M_{\chi^2}}(x) = N f(x) (F(x))^{N-1} \leq N \cdot \frac{1}{\sqrt{2\pi x}} \cdot \left(\frac{\sqrt{2x}}{\sqrt{\pi}} \right)^N = \frac{N}{\sqrt{2\pi}} \left(\frac{2}{\pi} \right)^{\frac{N-1}{2}} x^{\frac{N-2}{2}} \leq \frac{N}{\sqrt{2\pi}} \left(\frac{2}{\pi} \right)^{\frac{N-1}{2}}$$

For any constant $a \in (0, 1)$, we have the identity $\sum_{k=0}^{\infty} (k+1)a^k = (1-a)^{-2}$. Let $a = \sqrt{2/\pi}$, we can bound $N(\sqrt{2/\pi})^{N-1}$ via $Na^{N-1} \leq \sum_{k=N-1}^{\infty} (k+1)a^k \leq (1-a)^{-2}$. As a result,

$$\sup_{0 < t < 1} f_{M_{\chi^2}}(t) \leq \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{\left(1 - \sqrt{\frac{2}{\pi}}\right)^2} \quad (\text{S.6})$$

With (S.5) and (S.6), when $K = 1$,

$$\sup_t f_{M_{\chi^2}}(t) = \max \left\{ \sup_{0 < t < 1} f_{M_{\chi^2}}(t), \sup_{t \geq 1} f_{M_{\chi^2}}(t) \right\} \leq \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{\left(1 - \sqrt{\frac{2}{\pi}}\right)^2} < \infty. \quad (\text{S.7})$$

Next, we consider the case $K \geq 2$. Again, after rewriting $f_{M_{\chi^2}}(x)$ into the hazard form, we have (S.4). To bound $\sup_x f_{M_{\chi^2}}(x)$, it suffices to bound $\sup_x h(x)$. Note that χ_K^2 is essentially a $\text{Gamma}(\alpha, \theta)$ distribution with the shape $\alpha = \frac{K}{2}$ and scale $\theta = 2$, whose density function and survival function are

$$f(x) = \frac{1}{\Gamma(\alpha)\theta^\alpha} x^{\alpha-1} \exp\left(-\frac{x}{\theta}\right), \quad \bar{F}(x) = \int_x^\infty f(t) dt$$

Since the survival function of Gamma distribution admits a recursive property that

$$\bar{F}(x) = \int_x^\infty \frac{1}{\Gamma(\alpha)\theta^\alpha} x^{\alpha-1} \exp\left(-\frac{x}{\theta}\right) dt = \frac{\theta x^{\alpha-1} e^{-\frac{x}{\theta}}}{\Gamma(\alpha)\theta^\alpha} + \frac{\theta(\alpha-1)}{\Gamma(\alpha)\theta^\alpha} \int_x^\infty t^{\alpha-2} e^{-\frac{t}{\theta}} dt$$

By the non-negativity of the integral term,

$$\bar{F}(x) \geq \frac{\theta x^{\alpha-1} e^{-\frac{x}{\theta}}}{\Gamma(\alpha)\theta^\alpha} = \theta f(x).$$

Hence, for any $x > 0$,

$$h(x) = \frac{f(x)}{\bar{F}(x)} \leq \frac{1}{\theta} = \frac{1}{2}.$$

i.e., for all $K \geq 2$,

$$\sup_x f_{M_{\chi^2}}(x) \leq 2e^{-1} \sup_x h(x) \leq 2e^{-1} \cdot \frac{1}{2} = e^{-1} < \infty. \quad (\text{S.8})$$

(S.7) and (S.8) verify (S.3). Together with (S.2), we complete the proof of (S.1). $\square$

A.2. Proof of Proposition 3.5

Proof. Recall that, for each $i = 1, \dots, N$, W_i is the Wald statistic for testing

$$H_{0i} : \beta_i = (\beta_{i1}, \dots, \beta_{iK})^\top = \mathbf{0} \quad (\text{S.9})$$

in the linear panel model (1). Let $\mathbf{z}_{i,t} = (y_{t-1}, \dots, y_{t-K}, x_{t-1}, \dots, x_{t-K})^\top \in \mathbb{R}^{2K}$ and $\boldsymbol{\theta}_i = (\gamma_{i1}, \dots, \gamma_{iK}, \beta_{i1}, \dots, \beta_{iK})^\top \in \mathbb{R}^{2K}$. The model (1) becomes

$$y_{i,t} = \alpha_i + \mathbf{z}_{i,t}^\top \boldsymbol{\theta}_i + \epsilon_{i,t},$$

or equivalently,

$$\mathbf{y}_{i\cdot} = \alpha_i \mathbf{1}_T + \mathbf{Z}_i^\top \boldsymbol{\theta}_i + \boldsymbol{\epsilon}_{i\cdot},$$

where $\mathbf{y}_{i\cdot} = (y_{i,1}, \dots, y_{i,T})^\top \in \mathbb{R}^T$, $\mathbf{1}_T \in \mathbb{R}^T$ is a vector of 1s of length T , $\mathbf{Z}_i = (\mathbf{z}_{i,1}, \dots, \mathbf{z}_{i,T})^\top \in \mathbb{R}^{T \times 2K}$, and $\boldsymbol{\epsilon}_{i\cdot} = (\epsilon_{i,1}, \dots, \epsilon_{i,T})^\top \in \mathbb{R}^T$. Without loss of generality, we assume $\mathbf{Z}_i$ is centered satisfying $\mathbf{1}_T^\top \mathbf{Z}_i = \mathbf{0}$. The null hypothesis (S.9) becomes

$$H_{0i} : \mathbf{R}\boldsymbol{\theta}_i = \mathbf{0} \quad (\text{S.10})$$

with $\mathbf{R} = [\mathbf{0}_{K \times K}, \mathbf{I}_{K \times K}] \in \mathbb{R}^{K \times 2K}$. The OLS estimator of $\boldsymbol{\theta}_i$ can be obtained by $\hat{\boldsymbol{\theta}}_i = (\mathbf{Z}_i^\top \mathbf{Z}_i)^{-1} \mathbf{Z}_i^\top \mathbf{y}_{i\cdot}$, and the Wald statistic W_i takes the form of

$$\begin{aligned} W_i &= (\mathbf{R}\hat{\boldsymbol{\theta}}_i)^\top [\hat{\sigma}_i^2 \mathbf{R}(\mathbf{Z}_i^\top \mathbf{Z}_i)^{-1} \mathbf{R}^\top]^{-1} (\mathbf{R}\hat{\boldsymbol{\theta}}_i) \\ &= \boldsymbol{\epsilon}_{i\cdot}^\top \mathbf{Z}_i (\mathbf{Z}_i^\top \mathbf{Z}_i)^{-1} \mathbf{R}^\top [\hat{\sigma}_i^2 \mathbf{R}(\mathbf{Z}_i^\top \mathbf{Z}_i)^{-1} \mathbf{R}^\top]^{-1} \mathbf{R} (\mathbf{Z}_i^\top \mathbf{Z}_i)^{-1} \mathbf{Z}_i^\top \boldsymbol{\epsilon}_{i\cdot}. \end{aligned} \quad (\text{S.11})$$

Denote $\mathbf{Q}_i = \frac{1}{n} \mathbf{Z}_i^\top \mathbf{Z}_i$ and $\tilde{\mathbf{Q}}_i = \mathbb{E}(\mathbf{z}_{it} \mathbf{z}_{it}^\top)$, then the Wald statistic W_i becomes

$$W_i = \frac{1}{T \hat{\sigma}_i^2} \boldsymbol{\epsilon}_{i\cdot}^\top \mathbf{Z}_i \mathbf{Q}_i^{-1} \mathbf{R}^\top [\mathbf{R} \mathbf{Q}_i^{-1} \mathbf{R}^\top]^{-1} \mathbf{R} \mathbf{Q}_i^{-1} \mathbf{Z}_i^\top \boldsymbol{\epsilon}_{i\cdot}. \quad (\text{S.12})$$

Naturally, with Assumptions (A1)–(A4), W_i asymptotically follows a χ_K^2 distribution. We denote $W_i \xrightarrow{d} U_i \sim \chi_K^2$, where each U_i is marginally a χ_K^2 random variable, and the dependence among $(U_1, \dots, U_N)^\top$ resembles the dependence among $(W_1, \dots, W_N)^\top$. It suffices to show the Kolmogorov bound between $\max_{1 \leq i \leq N} W_i$ and $\max_{1 \leq i \leq N} U_i$. To this end, we further consider

$$W_i^{(0)} = \frac{1}{T \sigma_i^2} \boldsymbol{\epsilon}_{i\cdot}^\top \mathbf{Z}_i \mathbf{Q}_i^{-1} \mathbf{R}^\top [\mathbf{R} \mathbf{Q}_i^{-1} \mathbf{R}^\top]^{-1} \mathbf{R} \mathbf{Q}_i^{-1} \mathbf{Z}_i^\top \boldsymbol{\epsilon}_{i\cdot}. \quad (\text{S.13})$$

and

$$\tilde{W}_i = \frac{1}{T \sigma_i^2} \boldsymbol{\epsilon}_{i\cdot}^\top \mathbf{Z}_i \tilde{\mathbf{Q}}_i^{-1} \mathbf{R}^\top [\mathbf{R} \tilde{\mathbf{Q}}_i^{-1} \mathbf{R}^\top]^{-1} \mathbf{R} \tilde{\mathbf{Q}}_i^{-1} \mathbf{Z}_i^\top \boldsymbol{\epsilon}_{i\cdot}. \quad (\text{S.14})$$

Let $M_{Wald} = \max_{1 \leq i \leq N} W_i$, $M_{Wald}^{(0)} = \max_{1 \leq i \leq N} W_i^{(0)}$, $\tilde{M}_{Wald} = \max_{1 \leq i \leq N} \tilde{W}_i$, $M_{\chi_K^2} = \max_{1 \leq i \leq N} U_i$, and let $d(U, V) = \sup_x |\mathbb{P}(X \leq x) - \mathbb{P}(Y \leq x)|$ denote the Kolmogorov distance between the distributions of two random variables X and Y . Then,

$$\begin{aligned} d(M_{Wald}, M_{\chi_K^2}) &= \sup_x |\mathbb{P}(M_{Wald} \leq x) - \mathbb{P}(M_{\chi_K^2} \leq x)| \\ &\leq \sup_x |\mathbb{P}(\tilde{M}_{Wald} \leq x) - \mathbb{P}(M_{\chi_K^2} \leq x)| \\ &\quad + \sup_x |\mathbb{P}(M_{Wald}^{(0)} \leq x) - \mathbb{P}(\tilde{M}_{Wald} \leq x)| \\ &\quad + \sup_x |\mathbb{P}(M_{Wald} \leq x) - \mathbb{P}(M_{Wald}^{(0)} \leq x)| \\ &:= d(\tilde{M}_{Wald}, M_{\chi_K^2}) + d(M_{Wald}^{(0)}, \tilde{M}_{Wald}) + d(M_{Wald}, M_{Wald}^{(0)}) \end{aligned} \quad (\text{S.15})$$

where $d(M_{\chi_K^2}, \tilde{M}_{Wald})$ is the Gaussian approximation error, $d(\tilde{M}_{Wald}, M_{Wald}^{(0)})$ is the random design linearization error, and $d(M_{Wald}^{(0)}, M_{Wald})$ is the studentization error.

In what follows, we prove the non-asymptotic error bounds on the three error components in Lemma A.2, Lemma A.3, and Lemma A.4, respectively. Combining (S.16), (S.17), and (S.29), we obtain

$$\sup_{x>0} \left| \mathbb{P} \left(\max_{1 \leq i \leq N} W_i \geq x \right) - \mathbb{P} \left(\max_{1 \leq i \leq N} \chi_{K,i}^2 \geq x \right) \right| \lesssim \frac{\log^{7/6}(NT)}{T^{1/6}} + \frac{\log^{3/2} N}{\sqrt{T}} + \frac{\log^{7/4} N}{T^{5/4}} + \frac{1}{N^2}.$$

This completes the proof of Proposition 3.5. $\square$

Lemma A.2 (Non-asymptotic error bound on the Gaussian approximation error). *Under the same assumptions as in Proposition 3.5,*

$$d(\widetilde{M}_{Wald}, M_{\chi_K^2}) \lesssim \left(\frac{\log^7(NT)}{T} \right)^{1/6}. \quad (\text{S.16})$$

Proof of Lemma A.2. We first rewrite $\widetilde{W}_i$ into

$$\begin{aligned} \widetilde{W}_i &= \frac{1}{T\sigma_i^2} \boldsymbol{\varepsilon}_i^\top \mathbf{Z}_i \widetilde{\mathbf{Q}}_i^{-1} \mathbf{R}^\top [\mathbf{R} \widetilde{\mathbf{Q}}_i^{-1} \mathbf{R}^\top]^{-1} \mathbf{R} \widetilde{\mathbf{Q}}_i^{-1} \mathbf{Z}_i^\top \boldsymbol{\varepsilon}_i. \\ &= \left\| [\mathbf{R} \widetilde{\mathbf{Q}}_i^{-1} \mathbf{R}^\top]^{-1} \mathbf{R} \widetilde{\mathbf{Q}}_i^{-1} \frac{1}{\sqrt{T}\sigma_i} \sum_{t=1}^T \mathbf{z}_{i,t} \epsilon_{i,t} \right\|_2^2 := \left\| \frac{1}{\sqrt{T}} \sum_{t=1}^T \boldsymbol{\xi}_{i,t} \right\|_2^2 \end{aligned}$$

where $\boldsymbol{\xi}_{i,t} := \sigma_i^{-1} [\mathbf{R} \widetilde{\mathbf{Q}}_i^{-1} \mathbf{R}^\top]^{-1} \mathbf{R} \widetilde{\mathbf{Q}}_i^{-1} \mathbf{z}_{i,t} \epsilon_{i,t} \in \mathbb{R}^K$ with $\mathbb{E}(\boldsymbol{\xi}_{i,t}) = \mathbf{0}_K$ and $\text{cov}(\boldsymbol{\xi}_{i,t}) = \mathbf{I}_K$. Let $\boldsymbol{\xi}_{i,t}^G$ be the Gaussian counterpart of $\boldsymbol{\xi}_{i,t}$, satisfying $\boldsymbol{\xi}_{i,t}^G \sim \mathcal{N}(\mathbf{0}_K, \mathbf{I}_K)$, and therefore $\|\boldsymbol{\xi}_{i,t}^G\|_2^2 \sim \chi_K^2$. Let $\zeta_{i,t} = \mathbf{z}_{i,t}^\top \epsilon_{i,t} \in \mathbb{R}^{2K}$. Assumptions 3.1 and 3.4 imply that $\zeta_{i,t}$ follows a sub-exponential distribution, as a result, for any $k \in [2K]$, there exists $C > 0$, such that $\mathbb{E}(|\zeta_{i,t,k}|/C) \leq 2$, $i \in [N]$, $t \in [T]$. Hence, the high-dimensional central limit theorem (Theorem 2.1 and Proposition 2.1, Chernozhukov et al., 2017) implies,

$$\sup_x \left| \mathbb{P} \left(\max_{1 \leq i \leq N} \left\| \frac{1}{\sqrt{T}} \sum_{t=1}^T \boldsymbol{\xi}_{i,t} \right\|_2^2 \leq x \right) - \mathbb{P} \left(\max_{1 \leq i \leq N} \left\| \frac{1}{\sqrt{T}} \sum_{t=1}^T \boldsymbol{\xi}_{i,t}^G \right\|_2^2 \leq x \right) \right| \leq C \left(\frac{\log^7(NT)}{T} \right)^{1/6},$$

i.e.,

$$d(\widetilde{M}_{Wald}, M_{\chi_K^2}) = \sup_x |\mathbb{P}(\widetilde{M}_{Wald} \leq x) - \mathbb{P}(M_{\chi_K^2} \leq x)| \leq C \left(\frac{\log^7(NT)}{T} \right)^{1/6}.$$

This completes the proof of (S.16) in Lemma A.2. $\square$

Lemma A.3 (Non-asymptotic error bound on the random design linearization error). *Under the same assumptions as in Proposition 3.5,*

$$d(M_{Wald}^{(0)}, \widetilde{M}_{Wald}) \lesssim \frac{(\log N)^{3/2}}{\sqrt{T}} + \left(\frac{\log^7(NT)}{T} \right)^{1/6} + \frac{1}{N^2}. \quad (\text{S.17})$$

Proof of Lemma A.3. Note that, for any $x > 0$ and any $\varepsilon > 0$,

$$\begin{aligned} & \left| \mathbb{P} \left(\max_{1 \leq i \leq N} W_i^{(0)} \leq x \right) - \mathbb{P} \left(\max_{1 \leq i \leq N} \widetilde{W}_i \leq x \right) \right| \\ & \leq \mathbb{P} \left(\left| \max_{1 \leq i \leq N} \widetilde{W}_i - x \right| \leq \varepsilon \right) + \mathbb{P} \left(\left| \max_{1 \leq i \leq N} W_i^{(0)} - \max_{1 \leq i \leq N} \widetilde{W}_i \right| > \varepsilon \right) \\ & \leq \mathbb{P} \left(\left| \widetilde{M}_{Wald} - x \right| \leq \varepsilon \right) + \mathbb{P} \left(\max_{1 \leq i \leq N} |W_i^{(0)} - \widetilde{W}_i| > \varepsilon \right) \\ & \leq \mathbb{P} \left(\left| M_{\chi_K^2} - x \right| \leq \varepsilon \right) + 2 \cdot \sup_x |\mathbb{P}(\widetilde{M}_{Wald} \leq x) - \mathbb{P}(M_{\chi_K^2} \leq x)| + \mathbb{P} \left(\max_{1 \leq i \leq N} |W_i^{(0)} - \widetilde{W}_i| > \varepsilon \right). \end{aligned}$$

Hence, for any $\varepsilon > 0$,

$$\begin{aligned} d(M_{Wald}^{(0)}, \widetilde{M}_{Wald}) &= \sup_x \left| \mathbb{P} \left(\max_{1 \leq i \leq N} W_i^{(0)} \leq x \right) - \mathbb{P} \left(\max_{1 \leq i \leq N} \widetilde{W}_i \leq x \right) \right| \\ &\leq \sup_x \mathbb{P} \left(|M_{\chi_K^2} - x| \leq \varepsilon \right) + 2 \cdot d(M_{\chi_K^2}, \widetilde{M}_{Wald}) + \mathbb{P} \left(\max_{1 \leq i \leq N} |W_i^{(0)} - \widetilde{W}_i| > \varepsilon \right). \end{aligned} \quad (\text{S.18})$$

Let $h(\widetilde{\mathbf{Q}}_i) := \widetilde{\mathbf{Q}}_i^{-1} \mathbf{R}^\top [\mathbf{R} \widetilde{\mathbf{Q}}_i^{-1} \mathbf{R}^\top]^{-1} \mathbf{R} \widetilde{\mathbf{Q}}_i^{-1}$, $h(\mathbf{Q}_i) := \mathbf{Q}_i^{-1} \mathbf{R}^\top [\mathbf{R} \mathbf{Q}_i^{-1} \mathbf{R}^\top]^{-1} \mathbf{R} \mathbf{Q}_i^{-1}$, and $\mathbf{S}_i := \frac{1}{\sqrt{T} \sigma_i} \mathbf{Z}_i^\top \boldsymbol{\varepsilon}_i = \frac{1}{\sqrt{T} \sigma_i} \sum_{t=1}^T \mathbf{z}_{i,t} \epsilon_{i,t}$. The two statistics $\widetilde{W}_i$ and $W_i^{(0)}$ can therefore be rewritten as

$$W_i^{(0)} = \mathbf{S}_i^\top h(\mathbf{Q}_i) \mathbf{S}_i, \quad \widetilde{W}_i = \mathbf{S}_i^\top h(\widetilde{\mathbf{Q}}_i) \mathbf{S}_i$$

and their difference becomes

$$|W_i^{(0)} - \widetilde{W}_i| = \left| \mathbf{S}_i^\top \left(h(\mathbf{Q}_i) - h(\widetilde{\mathbf{Q}}_i) \right) \mathbf{S}_i \right| \leq \|h(\mathbf{Q}_i) - h(\widetilde{\mathbf{Q}}_i)\|_{op} \cdot \|\mathbf{S}_i\|_2^2 \quad (\text{S.19})$$

With Assumption (A3), for any $i \in [N]$,

$$\|h(\mathbf{Q}_i) - h(\widetilde{\mathbf{Q}}_i)\|_{op} \leq C \|\mathbf{Q}_i^{-1} - \widetilde{\mathbf{Q}}_i^{-1}\|_{op} \leq C \|\mathbf{Q}_i^{-1}\|_{op} \cdot \|\mathbf{Q}_i - \widetilde{\mathbf{Q}}_i\|_{op} \cdot \|\widetilde{\mathbf{Q}}_i^{-1}\|_{op} \leq C \|\mathbf{Q}_i - \widetilde{\mathbf{Q}}_i\|_{op}.$$

Along with (S.19), we have

$$\max_{1 \leq i \leq N} |W_i^{(0)} - \widetilde{W}_i| \leq C \max_{1 \leq i \leq N} \|\mathbf{Q}_i - \widetilde{\mathbf{Q}}_i\|_{op} \cdot \max_{1 \leq i \leq N} \|\mathbf{S}_i\|_2^2. \quad (\text{S.20})$$

Furthermore, for any $\varepsilon > 0$ and any $\eta > 0$,

$$\mathbb{P} \left(\max_{1 \leq i \leq N} |W_i^{(0)} - \widetilde{W}_i| > \varepsilon \right) \leq \mathbb{P} \left(\max_{1 \leq i \leq N} \|\mathbf{Q}_i - \widetilde{\mathbf{Q}}_i\|_{op} > \gamma \right) + \mathbb{P} \left(\max_{1 \leq i \leq N} \|\mathbf{S}_i\|_2^2 > \frac{\varepsilon}{C\eta} \right). \quad (\text{S.21})$$

We now focus on $\max_{1 \leq i \leq N} \|\mathbf{Q}_i - \widetilde{\mathbf{Q}}_i\|_{op}$. Recall that $\mathbf{Q}_i = \frac{1}{T} \mathbf{Z}_i^\top \mathbf{Z}_i \in \mathbb{R}^{2K \times 2K}$ and $\widetilde{\mathbf{Q}}_i = \mathbb{E}(\mathbf{z}_{i,t} \mathbf{z}_{i,t}^\top) \in \mathbb{R}^{2K \times 2K}$. With Assumptions 3.2 and 3.4, and using the matrix Bernstein inequality for sub-Gaussian sample covariance, there exists $C > 0$, such that

$$\mathbb{P} \left(\|\mathbf{Q}_i - \widetilde{\mathbf{Q}}_i\|_{op} \geq C \left(\sqrt{\frac{2K+u}{T}} + \frac{2K+u}{T} \right) \right) \leq 2e^{-u}.$$

To obtain the union bound, let $u = \log N + t$,

$$\begin{aligned} &\mathbb{P} \left(\max_{1 \leq i \leq N} \|\mathbf{Q}_i - \widetilde{\mathbf{Q}}_i\|_{op} \geq C \left(\sqrt{\frac{2K + \log N + t}{T}} + \frac{2K + \log N + t}{T} \right) \right) \\ &\leq \sum_{i=1}^N \mathbb{P} \left(\|\mathbf{Q}_i - \widetilde{\mathbf{Q}}_i\|_{op} \geq C \left(\sqrt{\frac{2K+u}{T}} + \frac{2K+u}{T} \right) \right) \leq N 2e^{-\log N + t} = 2e^{-t} \end{aligned} \quad (\text{S.22})$$

Next, we derive the concentration of $\|\mathbf{S}_i\|_2^2$. Recall that $\mathbf{S}_i := \frac{1}{\sqrt{T} \sigma_i} \mathbf{Z}_i^\top \boldsymbol{\varepsilon}_i = \frac{1}{\sqrt{T} \sigma_i} \sum_{t=1}^T \mathbf{z}_{i,t} \epsilon_{i,t}$. With Assumption 3.1 and Assumption 3.4, $\epsilon_{i,t}$ is sub-Gaussian with $\|\epsilon_{i,t}\|_{\psi_2} \leq K_\epsilon$ and $\mathbf{z}_{i,t}$ is sub-Gaussian with $\|\mathbf{z}_{i,t}\|_{\psi_2} \leq K_z$. Given the multiplicative nature of the elements in this statistic, we adopt a two-step approach by first studying a conditional tail bound given $\mathbf{Z}_i$, followed by an additional step to obtain the unconditional tail bound. By definition,

$$\|\mathbf{S}_i\|_2 = \sup_{\boldsymbol{\nu} \in \mathcal{S}^{2K-1}} \boldsymbol{\nu}^\top \mathbf{S}_i$$

where $\mathcal{S}^{2K-1} = \{\boldsymbol{\nu} \in \mathbb{R}^{2K} : \|\boldsymbol{\nu}\|_2 = 1\}$ is the unit sphere in $\mathbb{R}^{2K}$. For any individual unit vector $\boldsymbol{\nu} \in \mathcal{S}^{2K-1}$, $\boldsymbol{\nu}^\top \mathbf{S}_i = \frac{1}{\sqrt{T} \sigma_i} \sum_{t=1}^T \boldsymbol{\nu}^\top \mathbf{z}_{i,t} \epsilon_{i,t} := \frac{1}{\sqrt{T}} \sum_{t=1}^T \xi_{i,t}$ with $\xi_{i,t} := \sigma_i^{-1} \boldsymbol{\nu}^\top \mathbf{z}_{i,t} \epsilon_{i,t}$. Conditional on $\{\mathbf{z}_{i,t}\}_{t=1}^T$, each $\xi_{i,t}$ is

sub-Gaussian with $\|\xi_{i,t}\|_{\psi_2|Z} \leq \sigma_i^{-1} |\boldsymbol{\nu}^\top \mathbf{z}_{i,t}| \cdot \|\epsilon_{i,t}\|_{\psi_2} \leq \sigma_i^{-1} K_\epsilon |\boldsymbol{\nu}^\top \mathbf{z}_{i,t}|$, $\mathbb{E}(\xi_{i,t} | \mathbf{z}_{i,t}) = 0$, and $\{\xi_{i,t}\}_{t=1}^T$ are independent across t under conditioning. Hence, a conditional Bernstein-type bound yields, for any $u > 0$,

$$\mathbb{P} \left(\left| \sum_{t=1}^T \xi_{i,t} \right| \geq CK_\epsilon \sqrt{u} \left(\sum_{t=1}^T (\boldsymbol{\nu}^\top \mathbf{z}_{i,t})^2 \right)^{\frac{1}{2}} \mid \mathbf{Z}_i \right) \leq 2e^{-u}.$$

and equivalently,

$$\mathbb{P} \left(|\boldsymbol{\nu}^\top \mathbf{S}_i| \geq CK_\epsilon \frac{\sqrt{u}}{\sqrt{T}} \left(\sum_{t=1}^T (\boldsymbol{\nu}^\top \mathbf{z}_{i,t})^2 \right)^{\frac{1}{2}} \mid \mathbf{Z}_i \right) \leq 2e^{-u}.$$

Further note that $\sum_{t=1}^T (\boldsymbol{\nu}^\top \mathbf{z}_{i,t})^2 = \|\mathbf{Z}_i \boldsymbol{\nu}\|_2^2 \leq \|\mathbf{Z}_i\|_{op}^2 \cdot \|\boldsymbol{\nu}\|_2^2 = \|\mathbf{Z}_i\|_{op}^2$, then the above inequality becomes

$$\mathbb{P} \left(|\boldsymbol{\nu}^\top \mathbf{S}_i| \geq CK_\epsilon \|\mathbf{Z}_i\|_{op} \frac{\sqrt{u}}{\sqrt{T}} \mid \mathbf{Z}_i \right) \leq 2e^{-u}. \quad (\text{S.23})$$

Let $\mathcal{N}$ be a $\frac{1}{2}$ -net of $\mathcal{S}^{2K-1}$ in Euclidean norm with cardinality $|\mathcal{N}| \leq 5^{2K}$. By definition, $\|\mathbf{S}_i\|_2 = \sup_{\boldsymbol{\nu} \in \mathcal{S}^{2K-1}} \boldsymbol{\nu}^\top \mathbf{S}_i \leq 2 \max_{\boldsymbol{\nu} \in \mathcal{N}} \boldsymbol{\nu}^\top \mathbf{S}_i$. Applying the bound of each $\boldsymbol{\nu}$ and considering the union bound, conditional on $\mathbf{Z}_i$,

$$\mathbb{P} \left(\|\mathbf{S}_i\|_2 \geq 2CK_\epsilon \|\mathbf{Z}_i\|_{op} \frac{\sqrt{u}}{\sqrt{T}} \mid \mathbf{Z}_i \right) \leq \mathbb{P} \left(\max_{\boldsymbol{\nu} \in \mathcal{N}} |\boldsymbol{\nu}^\top \mathbf{S}_i| \geq CK_\epsilon \|\mathbf{Z}_i\|_{op} \frac{\sqrt{u}}{\sqrt{T}} \mid \mathbf{Z}_i \right) \leq |\mathcal{N}| \cdot 2e^{-u}.$$

Let $u = \log |\mathcal{N}| + t$, and note that $\log |\mathcal{N}| \asymp 2K$, the above becomes

$$\mathbb{P} \left(\|\mathbf{S}_i\|_2 \geq 2CK_\epsilon \|\mathbf{Z}_i\|_{op} \frac{\sqrt{2K+t}}{\sqrt{T}} \mid \mathbf{Z}_i \right) \leq 2e^{-t}. \quad (\text{S.24})$$

In addition, by the sub-Gaussianity of $\mathbf{z}_{i,t}$, for any $u > 0$,

$$\mathbb{P} \left(\|\mathbf{Z}_i\|_{op} \geq CK_z (\sqrt{T} + \sqrt{2K} + \sqrt{u}) \right) \leq 2e^{-u}. \quad (\text{S.25})$$

We then define two events,

$$\begin{aligned} A_{i,u} &:= \left\{ \|\mathbf{Z}_i\|_{op} \leq CK_z (\sqrt{T} + \sqrt{2K} + \sqrt{u}) \right\} \\ B_{i,u} &:= \left\{ \|\mathbf{S}_i\|_2 \leq 2CK_\epsilon \|\mathbf{Z}_i\|_{op} \frac{\sqrt{2K+u}}{\sqrt{T}} \right\} \end{aligned}$$

Previous results show that $\mathbb{P}(A_{i,u}^c) \leq 2e^{-u}$, $\mathbb{P}(B_{i,u}^c) \mid \mathbf{Z}_i \leq 2e^{-u}$, and hence $\mathbb{P}(B_{i,u}^c) \leq 2e^{-u}$ by taking expectations. On the event $A_{i,u} \cap B_{i,u}$,

$$\|\mathbf{S}_i\|_2 \leq CK_\epsilon K_z (\sqrt{T} + \sqrt{2K} + \sqrt{u}) \cdot \frac{\sqrt{2K+u}}{\sqrt{T}} \lesssim \frac{\sqrt{T} \cdot \sqrt{2K+u} + (2K+u)}{\sqrt{T}}$$

To get a cleaner form, with $2K+u \lesssim T$, we only keep the leading term, yielding

$$\mathbb{P} \left(\|\mathbf{S}_i\|_2 \geq CK_z K_\epsilon \cdot \frac{\sqrt{T \cdot 2K + T \cdot u}}{\sqrt{T}} \right) \leq 4e^{-u}$$

Then, the squaring brings

$$\mathbb{P} (\|\mathbf{S}_i\|_2^2 \geq C(2K+u)) \leq 4e^{-u}.$$

Let $u = \log N + t$, and we obtain the union bound

$$\mathbb{P} \left(\max_{1 \leq i \leq N} \|\mathbf{S}_i\|_2^2 \geq C(2K + \log N + t) \right) \leq N \cdot 4e^{-\log N + t} = 4e^{-t}. \quad (\text{S.26})$$

With (S.22) and (S.26), we have $\max_{1 \leq i \leq N} \|\mathbf{Q}_i - \tilde{\mathbf{Q}}_i\|_{op} = O_p(\sqrt{\frac{\log N}{T}})$ and $\|\mathbf{S}_i\|_2^2 = O_p(\log N)$. Then by (S.20), $\max_{1 \leq i \leq N} |W_i^{(0)} - \tilde{W}_i| = O_p(\frac{\log^{3/2} N}{\sqrt{T}})$. If we pick $t = 2 \log N$, (S.22) becomes

$$\mathbb{P} \left(\max_{1 \leq i \leq N} \|\mathbf{Q}_i - \tilde{\mathbf{Q}}_i\|_{op} \geq C \sqrt{\frac{\log N}{T}} \right) \leq 2e^{-2 \log N} = 2N^{-2},$$

and (S.26) becomes

$$\mathbb{P} \left(\max_{1 \leq i \leq N} \|\mathbf{S}_i\|_2^2 \geq C \log N \right) \leq 4e^{-2 \log N} = 4N^{-2}.$$

With these results, we choose $\varepsilon \asymp \frac{\log^{3/2} N}{\sqrt{T}}$ and $\eta \asymp \frac{\sqrt{\log N}}{\sqrt{T}}$ in (S.21), and hence $\frac{\varepsilon}{\eta} \asymp \log N$. The inequality becomes

$$\begin{aligned} & \mathbb{P} \left(\max_{1 \leq i \leq N} |W_i^{(0)} - \tilde{W}_i| > \frac{C \log^{3/2} N}{\sqrt{T}} \right) \\ & \leq \mathbb{P} \left(\max_{1 \leq i \leq N} \|\mathbf{Q}_i - \tilde{\mathbf{Q}}_i\|_{op} > C \frac{\sqrt{\log N}}{\sqrt{T}} \right) + \mathbb{P} \left(\max_{1 \leq i \leq N} \|\mathbf{S}_i\|_2^2 > C \log N \right) \lesssim N^{-2} \end{aligned} \quad (\text{S.27})$$

In addition, as proven in Lemma A.1, $\sup_x \mathbb{P}(|M_{\chi_K^2} - x| \leq \varepsilon) \lesssim \varepsilon \lesssim \frac{\log^{3/2} N}{\sqrt{T}}$, and (S.16) implies $d(M_{\chi_K^2}, \tilde{M}_{Wald}) \lesssim \left(\frac{\log^7(NT)}{T}\right)^{1/6}$. Then, by (S.18),

$$d(M_{Wald}^{(0)}, \tilde{M}_{Wald}) \lesssim \frac{(\log N)^{3/2}}{\sqrt{T}} + \left(\frac{\log^7(NT)}{T}\right)^{1/6} + \frac{1}{N^2} \quad (\text{S.28})$$

This completes the proof of (S.17) in Lemma A.3. $\square$

Lemma A.4 (Non-asymptotic error bound on the random design linearization error). *Under the same assumptions as in Proposition 3.5,*

$$d(M_{Wald}, M_{Wald}^{(0)}) \lesssim \frac{(\log N)^{7/4}}{T^{5/4}} + \frac{1}{N^2} + \left(\frac{\log^7(NT)}{T}\right)^{1/6} + \frac{(\log N)^{3/2}}{\sqrt{T}} \quad (\text{S.29})$$

Proof of Lemma A.4. Revisiting (S.12) and (S.13), by construction,

$$W_i = \frac{\sigma_i^2}{\hat{\sigma}_i^2} W_i^{(0)}$$

where $\sigma_i^2 = \mathbb{E}(\epsilon_{it}^2)$ and $\hat{\sigma}_i^2 = \hat{\boldsymbol{\varepsilon}}_i^\top \hat{\boldsymbol{\varepsilon}}_i / (T - 2K - 1)$, where $\hat{\boldsymbol{\varepsilon}}_i = \mathbf{y}_i - \hat{\alpha}_i \mathbf{1}_T - \mathbf{Z}_i^\top \hat{\boldsymbol{\theta}}_i$. With Assumption (A1), we know, for each $i \in [N]$,

$$\frac{\sigma_i^2}{\hat{\sigma}_i^2} - 1 = O_p\left(\frac{1}{\sqrt{T - 2K - 1}}\right).$$

Throughout our analysis, we consider K to be a fixed and small number that satisfies $K = o(T)$, and thus, we simplify $O_p((T - 2K - 1)^{-1/2})$ to $O_p(T^{-1/2})$. To translate the scaling error into the tail probability gap for the maxima, we consider a small positive error term $\delta_T \in (0, \frac{1}{2})$, and define an event $\mathcal{E}_\delta := \left\{ \max_{1 \leq i \leq N} \left| \frac{\sigma_i^2}{\hat{\sigma}_i^2} - 1 \right| \leq \delta \right\}$. On the event $\mathcal{E}_\delta$, we have

$$(1 - \delta) M_{Wald}^{(0)} \leq M_{Wald} \leq (1 + \delta) M_{Wald}^{(0)}.$$

Therefore, we can show

$$\mathbb{P}(M_{Wald} \leq x) \leq \mathbb{P}\left(M_{Wald}^{(0)} \leq \frac{x}{1 - \delta}\right) + \mathbb{P}\left(\max_{1 \leq i \leq N} \left| \frac{\sigma_i^2}{\hat{\sigma}_i^2} - 1 \right| > \delta\right),$$

and

$$\mathbb{P}(M_{Wald} \leq x) \geq \mathbb{P}\left(M_{Wald}^{(0)} \leq \frac{x}{1+\delta}\right) - \mathbb{P}\left(\max_{1 \leq i \leq N} \left| \frac{\sigma_i^2}{\hat{\sigma}_i^2} - 1 \right| > \delta\right).$$

With the above two equations,

$$\begin{aligned} d(M_{Wald}, M_{Wald}^{(0)}) &= \sup_x \left| \mathbb{P}(M_{Wald} \leq x) - \mathbb{P}(M_{Wald}^{(0)} \leq x) \right| \\ &\leq \max \left\{ \sup_x \left(\mathbb{P}(M_{Wald} \leq x) - \mathbb{P}(M_{Wald}^{(0)} \leq x) \right), \sup_x \left(\mathbb{P}(M_{Wald}^{(0)} \leq x) - \mathbb{P}(M_{Wald} \leq x) \right) \right\} \\ &\leq \sup_x \mathbb{P}\left(\frac{x}{1+\delta} < M_{Wald}^{(0)} \leq \frac{x}{1-\delta}\right) + \mathbb{P}\left(\max_{1 \leq i \leq N} \left| \frac{\sigma_i^2}{\hat{\sigma}_i^2} - 1 \right| > \delta\right). \end{aligned} \quad (\text{S.30})$$

With Assumption (A1), $\epsilon_{i,t}$ is sub-Gaussian. By the Hanson-Wright concentration inequality for quadratic forms (Theorem 1.1, [Rudelson & Vershynin, 2013](#)), there exists $c > 0$, such that for any $t > 0$,

$$\mathbb{P}\left(\left| \frac{\sigma_i^2}{\hat{\sigma}_i^2} - 1 \right| > t\right) \leq 2 \exp(-c(T - 2K - 1) \min\{t^2, t\}) \quad (\text{S.31})$$

Here, we consider a small $\delta \in (0, \frac{1}{2})$, leading to $\min\{\delta^2, \delta\} = \delta$. Furthermore, let $\delta = c_\delta \sqrt{\frac{\log N}{T - 2K - 1}} \asymp \sqrt{\frac{\log N}{T}}$, where c_δ satisfies $c_\delta > \sqrt{3c^{-1}}$. Hence,

$$\mathbb{P}\left(\max_{1 \leq i \leq N} \left| \frac{\sigma_i^2}{\hat{\sigma}_i^2} - 1 \right| > \delta\right) \leq N \cdot 2 \exp(-c \cdot c_\delta^2 \log N) = 2N^{-(c \cdot c_\delta^2 - 1)} \leq 2N^{-2}. \quad (\text{S.32})$$

To quantify $\sup_x \mathbb{P}\left(\frac{x}{1+\delta} < M_{Wald}^{(0)} \leq \frac{x}{1-\delta}\right)$, we consider the triangle inequality

$$\begin{aligned} &\sup_x \mathbb{P}\left(\frac{x}{1+\delta} < M_{Wald}^{(0)} \leq \frac{x}{1-\delta}\right) \\ &\leq \sup_x \mathbb{P}\left(\frac{x}{1+\delta} < M_{\chi^2} \leq \frac{x}{1-\delta}\right) + 2d(M_{Wald}^{(0)}, \widetilde{M}_{Wald}) + 2d(\widetilde{M}_{Wald}, M_{\chi^2}) \end{aligned} \quad (\text{S.33})$$

in which $d(M_{Wald}^{(0)}, \widetilde{M}_{Wald})$ has been established in (S.17), and $d(\widetilde{M}_{Wald}, M_{\chi^2})$ has been established in (S.16). It suffices to bound $\sup_x \mathbb{P}\left(\frac{x}{1+\delta} < M_{\chi^2} \leq \frac{x}{1-\delta}\right)$.

To obtain the anti-concentration property of $M_{\chi_K^2}$, recall $M_{\chi_K^2} = \max_{1 \leq i \leq N} U_i$ where $U_1, \dots, U_N \stackrel{i.i.d.}{\sim} \chi_K^2$. Rewriting U_i into $U_i = \|\mathbf{V}_i^G\|_2^2$, where $\mathbf{V}_i^G \sim \mathcal{N}(\mathbf{0}_K, \mathbf{I}_K)$. Then $M_{\chi_K^2} = \max_{1 \leq i \leq N} U_i = \max_{1 \leq i \leq N} \|\mathbf{V}_i^G\|_2^2 = (\max_{1 \leq i \leq N} \|\mathbf{V}_i^G\|_2)^2$. Let $T_G = \max_{1 \leq i \leq N} \|\mathbf{V}_i^G\|_2$, using the identity $\|\mathbf{V}_i^G\|_2 = \sup_{\boldsymbol{\nu} \in \mathcal{S}^{K-1}} \boldsymbol{\nu}^\top \mathbf{V}_i^G$, then

$$T_G = \max_{1 \leq i \leq N} \sup_{\boldsymbol{\nu} \in \mathcal{S}^{K-1}} \boldsymbol{\nu}^\top \mathbf{V}_i^G.$$

We approximate the sphere $\mathcal{S}^{K-1}$ by an τ -net $\mathcal{N}_\tau$ with cardinality $|\mathcal{N}_\tau| \leq (1 - 2\tau^{-1})^K$, then $\max_{1 \leq i \leq N} \max_{\boldsymbol{\nu} \in \mathcal{N}_\tau} \boldsymbol{\nu}^\top \mathbf{V}_i^G$ approximates T_G up to an additive error of the order $\tau \max_i \|\mathbf{V}_i^G\|_2$. With this, applying the Gaussian-max anti-concentration inequality (Theorem 3, [Chernozhukov et al., 2015](#)) to the discretized $\max_{1 \leq i \leq N} \max_{\boldsymbol{\nu} \in \mathcal{N}_\tau} \boldsymbol{\nu}^\top \mathbf{V}_i^G$ gives, for any $t > 0$,

$$\mathbb{P}(|T_G - t| \leq \Delta) \leq C\Delta \sqrt{\log(N \cdot |\mathcal{N}_\tau|)} \lesssim \Delta \sqrt{\log N}$$

For $x > 0$ and $0 < \Delta < x$,

$$\begin{aligned} &\mathbb{P}(|M_{\chi^2} - x| \leq \Delta) = \mathbb{P}(|T_G - \sqrt{x}| \leq \sqrt{x + \Delta} - \sqrt{x - \Delta}) \\ &\leq \mathbb{P}\left(|T_G - \sqrt{x}| \leq \frac{\Delta}{\sqrt{x}}\right) \leq \frac{C\Delta}{\sqrt{x}} \sqrt{\log N}. \end{aligned}$$

If we take $\Delta = x\delta$, the above becomes

$$\mathbb{P}(|M_{\chi^2} - x| \leq x\delta) \leq C\sqrt{x}\delta\sqrt{\log N}. \quad (\text{S.34})$$

It is worth mentioning that the above inequality is not a uniform bound. When taking supremum over x , the right-hand-side of (S.34) tends to infinity as $x \rightarrow \infty$. To obtain a uniform bound of $\sup_x \mathbb{P}\left(\frac{x}{1+\delta} < M_{\chi^2} \leq \frac{x}{1-\delta}\right)$ over x , we split the domain using a deterministic truncation bound B .

For the bounded region $\{0 < x \leq B\}$, note that for $\delta \in (0, \frac{1}{2})$,

$$\begin{aligned} \sup_{0 < x \leq B} \mathbb{P}\left(\frac{x}{1+\delta} < M_{\chi^2} \leq \frac{x}{1-\delta}\right) &\leq \sup_{0 < x \leq B} \mathbb{P}(|M_{\chi^2} - x| \leq 2\delta x) \\ &\leq \sup_{0 < x \leq B} C\delta\sqrt{x}\sqrt{\log N} \lesssim \delta\sqrt{B}\sqrt{\log N}, \end{aligned} \quad (\text{S.35})$$

in which the inequality of the 2nd line is a result of (S.34).

For the tail region $\{x > B\}$,

$$\sup_{x > B} \mathbb{P}\left(\frac{x}{1+\delta} < M_{\chi^2} \leq \frac{x}{1-\delta}\right) \leq \sup_{x > B} \mathbb{P}\left(M_{\chi^2} \geq \frac{x}{1+\delta}\right) \leq \mathbb{P}\left(M_{\chi^2} \geq \frac{B}{1+\delta}\right) \quad (\text{S.36})$$

For some positive constant $\gamma > 0$, we define

$$B := (1 + \delta)[K + 2\sqrt{3K \log N} + 6 \log N].$$

Asymptotically, $B \asymp \delta \log N$. Consistent with previous analysis, we consider $\delta \asymp \sqrt{\frac{\log N}{T}}$. With this choice of B , the rate in (S.35) becomes

$$\sup_{0 < x \leq B} \mathbb{P}\left(\frac{x}{1+\delta} < M_{\chi^2} \leq \frac{x}{1-\delta}\right) \lesssim \delta\sqrt{B}\sqrt{\log N} \asymp \frac{(\log N)^{7/4}}{T^{5/4}}. \quad (\text{S.37})$$

The inequality (S.36) becomes

$$\begin{aligned} \sup_{x > B} \mathbb{P}\left(\frac{x}{1+\delta} < M_{\chi^2} \leq \frac{x}{1-\delta}\right) &\leq \mathbb{P}\left(M_{\chi^2} \geq \frac{B}{1+\delta}\right) \\ &= \mathbb{P}\left(M_{\chi^2} \geq K + 2\sqrt{3K \log N} + 6 \log N\right) \leq Ne^{-3 \log N} = N^{-2} \end{aligned} \quad (\text{S.38})$$

With (S.37) and (S.38),

$$\begin{aligned} &\sup_x \mathbb{P}\left(\frac{x}{1+\delta} < M_{\chi^2} \leq \frac{x}{1-\delta}\right) \\ &= \max \left\{ \sup_{0 < x \leq B} \mathbb{P}\left(\frac{x}{1+\delta} < M_{\chi^2} \leq \frac{x}{1-\delta}\right), \sup_{x > B} \mathbb{P}\left(\frac{x}{1+\delta} < M_{\chi^2} \leq \frac{x}{1-\delta}\right) \right\} \\ &\lesssim \max \left\{ \frac{(\log N)^{7/4}}{T^{5/4}}, N^{-2} \right\} \lesssim \frac{(\log N)^{7/4}}{T^{5/4}} + N^{-2}. \end{aligned} \quad (\text{S.39})$$

With (S.32), (S.33), (S.39), (S.30) becomes

$$d(M_{Wald}, M_{Wald}^{(0)}) \lesssim \frac{(\log N)^{7/4}}{T^{5/4}} + \frac{1}{N^2} + \left(\frac{\log^7(NT)}{T}\right)^{1/6} + \frac{(\log N)^{3/2}}{\sqrt{T}} \quad (\text{S.40})$$

This completes the proof of (S.29) in Lemma A.4. $\square$

A.3. Proof of Proposition 3.7

Proof. Let $U \sim \chi_K^2$ be any random variable following a χ_K^2 distribution, and note that, by construction, we have $c(T) \geq 1.5$. Therefore,

$$\begin{aligned} \mathbb{P}(U \geq \delta_{N,T}) &= \mathbb{P}(U \geq (K + 2\sqrt{K \log N} + 2 \log N) \cdot c(T)) \\ &\leq \mathbb{P}(U \geq K + 2\sqrt{K \log N \cdot c(T)} + 2 \log N \cdot c(T)) \end{aligned} \quad (\text{S.41})$$

By Lemma 1 of [Laurent & Massart \(2000\)](#),

$$(\text{S.41}) \leq \exp(-\log(N \cdot c(T))) = N^{-c(T)}. \quad (\text{S.42})$$

For $U_1, \dots, U_N \sim \chi_K^2$, by the union bound,

$$\mathbb{P}(\max_{1 \leq i \leq N} U_i \geq \delta_{N,T}) \leq N \cdot \max_{1 \leq i \leq N} \mathbb{P}(U_i \geq \delta_{N,T}) \leq N^{1-c(T)}.$$

As a result,

$$\begin{aligned} &\mathbb{P}(\max_{1 \leq i \leq N} W_i \geq \delta_{N,T}) \\ &\leq \mathbb{P}(\max_{1 \leq i \leq N} U_i \geq \delta_{N,T}) + |\mathbb{P}(\max_{1 \leq i \leq N} W_i \geq \delta_{N,T}) - \mathbb{P}(\max_{1 \leq i \leq N} U_i \geq \delta_{N,T})| \\ &\leq N^{1-c(T)} + \sup_{x>0} \left| \mathbb{P}\left(\max_{1 \leq i \leq N} W_i \geq x\right) - \mathbb{P}\left(\max_{1 \leq i \leq N} \chi_{K,i}^2 \geq x\right) \right| \end{aligned} \quad (\text{S.43})$$

Proposition 3.5 shows that, under the null hypothesis, we have

$$d(\max_{1 \leq i \leq N} W_i, \max_{1 \leq i \leq N} \chi_{K,i}^2) \lesssim \left(\frac{\log^7(NT)}{T} \right)^{1/6} + \frac{(\log N)^{3/2}}{\sqrt{T}} + \frac{(\log N)^{7/4}}{T^{5/4}} + \frac{1}{N^2}. \quad (\text{S.44})$$

By construction, $N^{1-c(T)} \lesssim N^{-0.5}$. (S.43) becomes,

$$\mathbb{P}(\max_{1 \leq i \leq N} W_i \geq \delta_{N,T}) \leq \frac{1}{\sqrt{N}} + \left(\frac{\log^7(NT)}{T} \right)^{1/6} + \frac{(\log N)^{3/2}}{\sqrt{T}} + \frac{(\log N)^{7/4}}{T^{5/4}} \quad (\text{S.45})$$

This completes the proof of Proposition 3.7. $\square$

A.4. Proofs of Theorem 3.8

Proof. Property (i). By construction, $J_{PE} = \sqrt{N} \cdot M_{Wald} \cdot I\{M_{Wald} > \delta_{N,T}\} \geq 0$, and therefore, property (i) naturally holds.

Property (ii). By Proposition 3.7, under the null hypothesis,

$$\mathbb{P}\left(\max_{1 \leq i \leq N} W_i \geq \delta_{N,T}\right) \leq \frac{1}{\sqrt{N}} + \left(\frac{\log^7(NT)}{T} \right)^{1/6} + \frac{(\log N)^{3/2}}{\sqrt{T}} + \frac{(\log N)^{7/4}}{T^{5/4}} \quad (\text{S.46})$$

Suppose $\log N = o(T^{1/7})$, then

$$\begin{aligned} \frac{\log^7(NT)}{T} &= \frac{\log^7 N}{T} + \frac{\log^7 T}{T} = o(1) \\ \frac{(\log N)^{3/2}}{\sqrt{T}} &= \left(\frac{\log^7 N}{T} \right)^{1/2} \cdot \frac{1}{\log^2 N} = o(1) \\ \frac{(\log N)^{7/4}}{T^{5/4}} &= \left(\frac{\log^7 N}{T} \right)^{1/4} \cdot \frac{1}{T} = o(1). \end{aligned}$$

Hence, as $N, T \rightarrow \infty$,

$$\mathbb{P}(J_{PE} = 0 \mid H_0) = \mathbb{P}\left(\max_{1 \leq i \leq N} W_i \geq \delta_{N,T} \mid H_0\right) \rightarrow 0.$$

Property (iii). With Assumptions 3.1 – 3.3, under the null hypothesis $H_{0i} : \beta_i = \mathbf{0}$, $W_i \xrightarrow{d} \chi_K^2$ as $T \rightarrow \infty$. Under the alternative hypothesis $H_{ai} : \beta_i \neq \mathbf{0}$, W_i converges in distribution to a non-central χ^2 distribution $\chi_K^2(\lambda_i)$ with the non-centrality parameter $\lambda_i = T\beta_i^\top \Sigma_i^{-1} \beta_i$, where $\Sigma_i = T\sigma_i^2 \mathbf{R}^\top (\mathbf{Z}_i^\top \mathbf{Z}_i)^{-1} \mathbf{R}$, yielding

$$\mathbb{P}_{\beta_i}(W_i \geq \delta_{N,T}) = \mathbb{P}(\chi_K^2(\lambda_i) \geq \delta_{N,T}) + o(1).$$

To analyze the asymptotic power, we consider a widely used normal approximation for non-central $\chi_K^2(\lambda_i)$: $\chi_K^2(\lambda_i) \approx \mathcal{N}(K + \lambda_i, 2(K + 2\lambda_i))$, and with Berry-Esseen-type bound, we have

$$\mathbb{P}(\chi_K^2(\lambda_i) \geq \delta_{N,T}) = 1 - \Phi\left(\frac{\delta_{N,T} - (K + \lambda_i)}{\sqrt{2(K + 2\lambda_i)}}\right) + O(\lambda_i^{-\frac{1}{2}}) \quad (\text{S.47})$$

For any $\beta \in \Theta_\beta = \left\{(\beta_1, \dots, \beta_N) : \max_{1 \leq i \leq N} \beta_i^\top \Sigma_i^{-1} \beta_i > \frac{\eta \cdot \log \log T \cdot \log N}{T} \text{ for some } \eta > 2\right\}$, there exists $i_0 \in [N]$, such that $\lambda_{i_0} = T\beta_{i_0}^\top \Sigma_{i_0}^{-1} \beta_{i_0} > \eta \cdot \log \log T \cdot \log N$ for some $\eta > 2$. As a result,

$$\begin{aligned} \frac{\delta_{N,T} - (K + \lambda_i)}{\sqrt{2(K + 2\lambda_i)}} &\leq \frac{(K + 2\sqrt{K \log N} + 2 \log N) \cdot c(T) - (K + \eta \log N \log \log T)}{\sqrt{2(K + 2\eta \log N \log \log T)}} \\ &= \frac{K(c(T) - 1) + 2c(T)\sqrt{K \log N} + (2c(T) - \eta \log \log T) \log N}{\sqrt{2(K + 2\eta \log N \log \log T)}} \\ &\asymp \frac{(2c(T) - \eta \log \log T) \log N}{\sqrt{\log N \log \log T}} \rightarrow -\infty \quad \text{as } N, T \rightarrow \infty \end{aligned}$$

and therefore,

$$\Phi\left(\frac{\delta_{N,T} - (K + \lambda_i)}{\sqrt{2(K + 2\lambda_i)}}\right) \rightarrow 0 \quad \text{as } N, T \rightarrow \infty.$$

Then,

$$\mathbb{P}_{\beta_i}(W_{i_0} \geq \delta_{N,T}) = \mathbb{P}(\chi_K^2(\lambda_i) \geq \delta_{N,T}) + o(1) = 1 - \Phi\left(\frac{\delta_{N,T} - (K + \lambda_i)}{\sqrt{2(K + 2\lambda_i)}}\right) + o(1),$$

and hence,

$$\mathbb{P}_{\beta_i}(W_{i_0} \geq \delta_{N,T}) \rightarrow 1 \quad \text{as } N, T \rightarrow \infty.$$

For any $\beta \in \Theta_\beta$,

$$\mathbb{P}_\beta(J_{PE} \geq \sqrt{T}\delta_{N,T}) = \mathbb{P}_\beta(M_{Wald} > \delta_{N,T}) \geq \mathbb{P}_{\beta_i}(W_{i_0} \geq \delta_{N,T}) \rightarrow 1 \quad \text{as } N, T \rightarrow \infty.$$

□

A.5. Proofs of Theorem 3.9

Proof. By definition, $T_{PE-PGCT} = T_{pivotal} + J_{PE}$. Then, for any x ,

$$\begin{aligned} &\mathbb{P}_{H_0}(T_{pivotal} \leq x) \\ &= \mathbb{P}_{H_0}(\{T_{pivotal} \leq x\} \cap \{J_{PE} = 0\}) + \mathbb{P}_{H_0}(\{T_{pivotal} \leq x\} \cap \{J_{PE} \neq 0\}) \\ &\leq \mathbb{P}_{H_0}(\{T_{pivotal} + J_{PE} \leq x\} \cap \{J_{PE} = 0\}) + \mathbb{P}_{H_0}(J_{PE} \neq 0) \\ &\leq \mathbb{P}_{H_0}(T_{PE-PGCT} \leq x) + \mathbb{P}_{H_0}(J_{PE} \neq 0). \end{aligned} \quad (\text{S.48})$$

On the other hand, the non-negativity of J_{PE} ensures

$$\mathbb{P}_{H_0}(T_{PE-PGCT} \leq x) \leq \mathbb{P}_{H_0}(T_{pivotal} \leq x).$$

Therefore, for any x ,

$$0 \leq \mathbb{P}_{H_0}(T_{pivotal} \leq x) - \mathbb{P}_{H_0}(T_{PE-PGCT} \leq x) \leq \mathbb{P}_{H_0}(J_{PE} \neq 0).$$

Property (ii) of Theorem 3.8 suggests, $\mathbb{P}_{H_0}(J_{PE} \neq 0) \rightarrow 0$ as $N, T \rightarrow \infty$, and hence, for any x ,

$$\mathbb{P}_{H_0}(T_{PE-PGCT} \leq x) - \mathbb{P}_{H_0}(T_{pivotal} \leq x) \rightarrow 0,$$

i.e., under H_0 ,

$$T_{PE-PGCT} \stackrel{d}{=} T_{pivotal}.$$

□

A.6. Proofs of Theorem 3.10

Proof. Property (i). By Theorem 3.9, $T_{PE-PGCT} \stackrel{d}{=} T_{pivotal}$ under H_0 . Therefore, for any $\beta \in \Theta_0 = \{(\beta_1, \dots, \beta_N) : \beta_i = \mathbf{0}, \forall i = 1, \dots, N\}$, as $N, T \rightarrow \infty$,

$$\phi_{PE-PGCT}(\beta = \mathbf{0}) = \mathbb{P}_{H_0}(T_{PE-PGCT} \geq q_{pivotal}^\alpha) \rightarrow \mathbb{P}_{H_0}(T_{pivotal} \geq q_{pivotal}^\alpha) = \alpha.$$

Property (ii). The non-negativity of J_{PE} ensures that $\mathbb{P}(T_{PE-PGCT} \geq x) \leq \mathbb{P}(T_{pivotal} \geq x)$. Therefore, for any $\beta \in \Theta$,

$$\phi_{PE-PGCT}(\beta) = \mathbb{P}_\beta(T_{PE-PGCT} \geq q_{pivotal}^\alpha) \geq \mathbb{P}_\beta(T_{pivotal} \geq q_{pivotal}^\alpha) = \phi_{pivotal}(\beta).$$

Property (iii). This property is a natural result of Property (iii) in Theorem 3.8. Specifically, Theorem 3.8 shows that $\forall \beta \in \Theta_\beta$, $J_{PE} \rightarrow \infty$ as $N, T \rightarrow \infty$. Hence, for $\forall \beta \in \Theta_\beta$, with the fixed critical value $q_{pivotal}^\alpha$, as $N, T \rightarrow \infty$,

$$\phi_{PE-PGCT}(\beta) = \mathbb{P}_\beta(T_{pivotal} + J_{PE} \geq q_{pivotal}^\alpha) \rightarrow 1.$$

□

B. Supplemental Details of the Pivotal Test $T_{pivotal}$

As detailed in Remark 2.2 of the main context, we advocate using a data-adaptive strategy and recommend different pivotal tests depending on the dimensionality of the panel. This data-adaptive choice leverages existing methods in the literature within their respective regimes of strength, thereby ensuring robustness and validity of the proposed procedure across a broad range of panel settings. Specifically, we propose using

$$T_{pivotal} = \begin{cases} |T_{DH}|, & \text{if } N < T - K \\ T_{HPJ}, & \text{if } N \geq T - K \end{cases}$$

where the first regime corresponds to the traditional panel setting and the second corresponds to the high-dimensional panel regime. The above choice is proposed after carefully weighing the relative advantages and limitations of available approaches in the literature. As empirically demonstrated in Figure 1, DH performs well when T is large but suffers from severe size distortions when N is large, particularly when $N > T$. In contrast, HPJ performs well for large N but exhibits distorted Type I error control when N is small. These differences are rooted in the distinct asymptotic regimes under which the two tests are valid. For completion, in what follows, we present the mathematical details of the two existing PGCT tests.

B.1. The Dumitrescu-Hurlin (DH) Test

The DH test (Dumitrescu & Hurlin, 2012) begins with the individual Wald statistics W_i for testing $H_{0i} : \beta_i = (\beta_{i1}, \dots, \beta_{iK})^\top = \mathbf{0}$ in the linear panel model (1), for each $i \in [N]$. The explicit form of W_i is given in (S.12). To conduct the Granger-causality test, the DH test considers the standardized average of individual Wald statistics, defined by

$$Z_{DH} = \sqrt{\frac{N}{2K}} \left(\frac{1}{N} \sum_{i=1}^N W_i - K \right).$$

Theorem 1 of [Dumitrescu & Hurlin \(2012\)](#) proves that under some regularity conditions, $Z_{DH} \xrightarrow{d} N(0, 1)$ in a sequential asymptotic regime when $T \rightarrow \infty$ first, followed by $N \rightarrow \infty$. To improve the finite-sample performance, [Dumitrescu & Hurlin \(2012\)](#) also introduces a fixed- T version, for any fixed $T > 5 + 2K$,

$$T_{DH} = \sqrt{\frac{p}{2K} \cdot \frac{T-2K-5}{T-K-3}} \left(\frac{T-2K-3}{T-2K-1} \cdot \frac{1}{N} \sum_{i=1}^N W_i - K \right).$$

Proposition 3 of [Dumitrescu & Hurlin \(2012\)](#) proves that under some regularity conditions, $T_{DH} \xrightarrow{d} N(0, 1)$ as $N \rightarrow \infty$. According to the analysis of limiting null distributions, the DH tests rejects H_0 at the significance level α if $|Z_{DH}| \geq z_{\alpha/2}$, or if using the fixed- T version for better finite-sample performance, the DH tests rejects H_0 at the significance level α if $|T_{DH}| \geq z_{\alpha/2}$, where $z_{\alpha/2}$ is the upper- $\alpha/2$ quantile of the standard normal distribution.

B.2. The Half-Panel Jackknife (HPJ) Test

The HPJ test ([Juodis et al., 2021](#)) employs a pooled least-squares estimator, that relies on a key insight that under the null hypothesis of non-causality, the Granger-causal coefficients are identically zero for all units, rendering the parameter of interest homogeneous under the null hypothesis. Assuming homogeneity in β_i , the model (1) becomes

$$\mathbf{y}_{i\cdot} = \Xi_i^\top \begin{pmatrix} \alpha_i \\ \gamma_i \end{pmatrix} + \mathbf{X}_i^\top \beta + \varepsilon_{i\cdot},$$

where $\mathbf{y}_{i\cdot} = (y_{i,1}, \dots, y_{i,T})^\top \in \mathbb{R}^T$, $\mathbf{X}_i = (\mathbf{x}_{i,1}, \dots, \mathbf{x}_{i,T})^\top \in \mathbb{R}^{T \times K}$, $\Xi_i = (\xi_{i,1}, \dots, \xi_{i,T})^\top \in \mathbb{R}^{T \times (K+1)}$ with $\xi_{i,t} = (1, y_{i,t-1}, \dots, y_{i,t-K})^\top \in \mathbb{R}^{K+1}$. The least-square estimator of β can be obtained by $\hat{\beta}_0 = (\sum_{i=1}^N \mathbf{X}_i^\top \mathbf{M}_i \mathbf{X}_i)^{-1} (\sum_{i=1}^N \mathbf{X}_i^\top \mathbf{M}_i \mathbf{y}_{i\cdot})$, where $\mathbf{M}_i = \mathbf{I}_T - \Xi_i (\Xi_i^\top \Xi_i)^{-1} \Xi_i^\top$. To remove the bias in dynamic panel settings, [Juodis et al. \(2021\)](#) adopts the Split-Panel Jackknife strategy ([Dhaene & Jochmans, 2015](#)) and considers

$$\hat{\beta} = 2\hat{\beta}_0 - \frac{\hat{\beta}_1 + \hat{\beta}_2}{2},$$

where $\hat{\beta}_1$ and $\hat{\beta}_2$ are the least-square estimators of β using the first $T_1 = T/2$ observations and the last $T_2 = T - T_1$ observations, respectively. The HPJ statistic is constructed using the quadratic form of the estimator $\hat{\beta}$, i.e.,

$$T_{HPJ} = \hat{\beta}^\top [\widehat{cov}(\hat{\beta})]^{-1} \hat{\beta},$$

where $\widehat{cov}(\hat{\beta})$ is an estimate of $cov(\hat{\beta})$, taking the explicit form of $\widehat{cov}(\hat{\beta}) = \frac{1}{NT} \hat{\mathbf{J}}^{-1} \hat{\mathbf{V}} \hat{\mathbf{J}}^{-1}$, with $\hat{\mathbf{J}} = \frac{1}{NT} \sum_{i=1}^N \mathbf{X}_i^\top \mathbf{M}_i \mathbf{X}_i$ and $\hat{\mathbf{V}} = \frac{1}{N(T-K-1)-K} \sum_{i=1}^N \mathbf{X}_i^\top \mathbf{M}_i (\mathbf{y}_{i\cdot} - \mathbf{X}_i \hat{\beta}_0)(\mathbf{y}_{i\cdot} - \mathbf{X}_i \hat{\beta}_0)^\top \mathbf{M}_i \mathbf{X}_i$.

With Theorem 2 of [Fernández-Val & Lee \(2013\)](#) and Corollary 3.1 of [Juodis et al. \(2021\)](#), under some regularity conditions, $T_{HPJ} \rightarrow \chi_K^2$ as $T, N \rightarrow \infty$. The HPJ rejects H_0 if $T_{HPJ} \geq q_\alpha$, where q_α is the upper- α quantile of the χ_K^2 distribution.

C. Supplemental Simulation Results

C.1. Simulation Results with non-Gaussian Errors

Besides the Gaussian error setting, we conduct simulations with non-Gaussian innovations. Specifically, errors are generated by $\varepsilon_{i,t} \sim$ standardized t_3 . Table S.1 summarizes the empirical Type-I error under the non-Gaussian error setting. The results are similar to those of the Gaussian error, demonstrating the performance of J_{PE} in accommodating non-Gaussian errors. For power evaluation, we again consider situations with different signal sparsity levels and different signal strengths. We adopt the same data-generating process as those in Section 4. Again, since DH and HPJ are known to have (sometimes severe) size distortions under certain configurations, we report size-corrected power for these two to ensure fair comparisons. Figure S.1 plots a representative subset of results under sparse alternatives, with complete results reported in Table S.4. Figure S.2 visualizes a representative subset of power results under various signal sparsity levels, with complete results reported in Table S.5. The results under the non-Gaussian setting largely resemble the patterns of the Gaussian setting discussed in Section 4.

Table S.1. Empirical Type I error under the non-Gaussian error setting (nominal level $\alpha = 0.05$).

T	N	PE-PGCT	DH	HPJ	T	N	PE-PGCT	DH	HPJ	T	N	PE-PGCT	DH	HPJ
50	5	0.059	0.058	0.158	100	5	0.046	0.046	0.156	200	5	0.058	0.058	0.154
	10	0.047	0.047	0.105		10	0.051	0.051	0.104		10	0.049	0.049	0.107
	20	0.070	0.070	0.087		20	0.051	0.049	0.075		20	0.038	0.038	0.071
	50	0.071	0.079	0.066		50	0.074	0.073	0.054		50	0.045	0.045	0.061
	100	0.065	0.079	0.062		100	0.061	0.054	0.060		100	0.055	0.055	0.045
	200	0.056	0.103	0.054		200	0.057	0.071	0.057		200	0.052	0.063	0.052
100	5	0.059	0.204	0.057	200	5	0.057	0.093	0.057	500	5	0.052	0.071	0.052
	10					10					10			
	20					20					20			
	50					50					50			
	100					100					100			
	200					200					200			

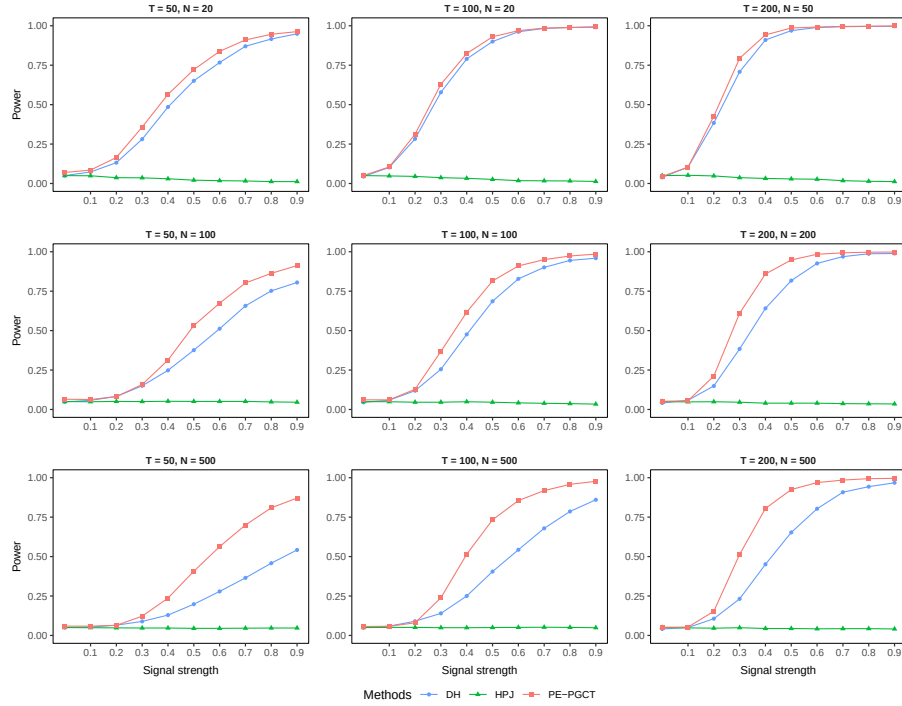


Figure S.1. Empirical power under sparse alternatives ($N_0 = 1$) under the non-Gaussian setting. Tests are conducted at level $\alpha = 0.05$.

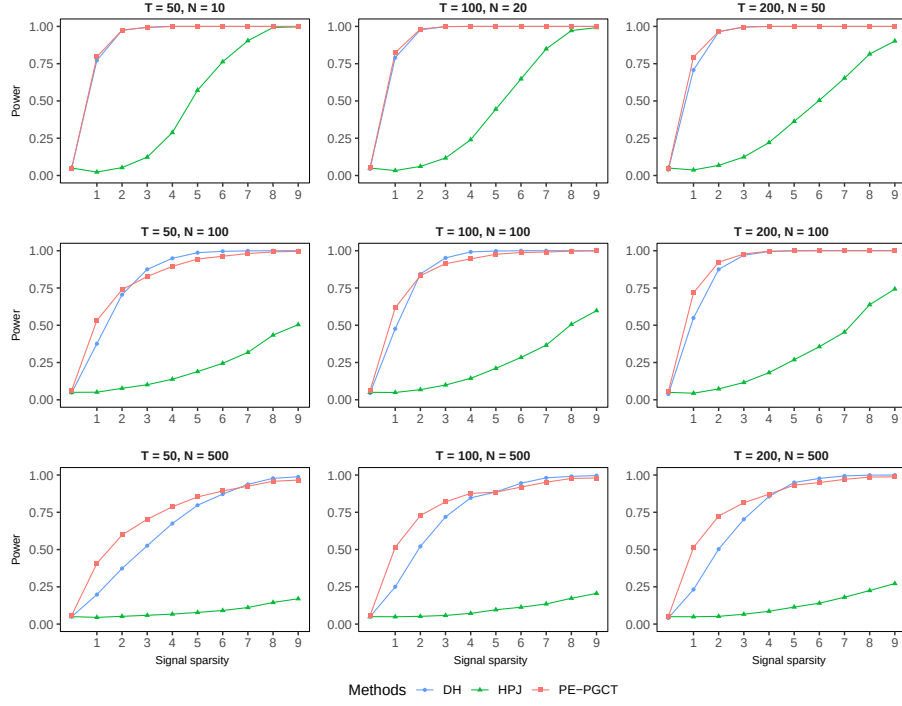


Figure S.2. Empirical power in situations of different sparsity levels (signal number ranging from 0 to 9) under the non-Gaussian setting. Tests are conducted at the nominal level $\alpha = 0.05$.

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Table S.2. Empirical power under sparse alternatives ($N_0 = 1$) under the Gaussian error setting. Tests are conducted at level $\alpha = 0.05$.

T	N	β	PE-PGCT	DH	HPJ	T	N	β	PE-PGCT	DH	HPJ	T	N	β	PE-PGCT	DH	HPJ
50	5	0	0.060	0.050	0.050	100	5	0	0.044	0.050	0.050	200	5	0	0.061	0.050	0.050
		0.1	0.089	0.082	0.049			0.1	0.136	0.147	0.050			0.1	0.278	0.261	0.037
		0.2	0.230	0.205	0.042			0.2	0.415	0.434	0.034			0.2	0.741	0.732	0.016
		0.3	0.428	0.408	0.028			0.3	0.760	0.777	0.012			0.3	0.959	0.954	0.005
		0.4	0.650	0.627	0.022			0.4	0.928	0.932	0.002			0.4	0.996	0.995	0.002
		0.5	0.813	0.795	0.015			0.5	0.985	0.988	0.001			0.5	0.999	0.999	0.000
		0.6	0.915	0.904	0.004			0.6	0.998	0.999	0.000			0.6	1.000	1.000	0.000
		0.7	0.953	0.950	0.001			0.7	1.000	1.000	0.000			0.7	1.000	1.000	0.000
		0.8	0.977	0.975	0.000			0.8	1.000	1.000	0.000			0.8	1.000	1.000	0.000
0.9	0.986	0.985	0.000	0.9	1.000	1.000	0.000	0.9	1.000	1.000	0.000						
50	10	0	0.052	0.050	0.050	100	10	0	0.041	0.049	0.050	200	10	0	0.052	0.049	0.050
		0.1	0.079	0.078	0.064			0.1	0.102	0.111	0.050			0.1	0.172	0.170	0.061
		0.2	0.173	0.167	0.062			0.2	0.334	0.349	0.036			0.2	0.606	0.599	0.045
		0.3	0.353	0.330	0.048			0.3	0.679	0.681	0.023			0.3	0.904	0.895	0.024
		0.4	0.560	0.534	0.033			0.4	0.873	0.870	0.013			0.4	0.989	0.987	0.008
		0.5	0.742	0.714	0.025			0.5	0.958	0.954	0.005			0.5	0.999	0.999	0.003
		0.6	0.863	0.846	0.018			0.6	0.986	0.985	0.002			0.6	1.000	1.000	0.001
		0.7	0.934	0.921	0.014			0.7	0.997	0.997	0.002			0.7	1.000	1.000	0.001
		0.8	0.965	0.954	0.009			0.8	1.000	0.999	0.001			0.8	1.000	1.000	0.000
0.9	0.987	0.980	0.006	0.9	1.000	1.000	0.000	0.9	1.000	1.000	0.000						
50	20	0	0.062	0.049	0.050	100	20	0	0.056	0.047	0.050	200	20	0	0.060	0.047	0.050
		0.1	0.088	0.074	0.047			0.1	0.086	0.078	0.057			0.1	0.133	0.118	0.060
		0.2	0.159	0.137	0.044			0.2	0.268	0.242	0.053			0.2	0.526	0.480	0.047
		0.3	0.312	0.271	0.044			0.3	0.554	0.507	0.044			0.3	0.860	0.831	0.031
		0.4	0.494	0.426	0.040			0.4	0.793	0.754	0.032			0.4	0.974	0.960	0.018
		0.5	0.666	0.606	0.036			0.5	0.925	0.892	0.024			0.5	0.998	0.994	0.009
		0.6	0.800	0.733	0.030			0.6	0.981	0.965	0.017			0.6	1.000	1.000	0.005
		0.7	0.880	0.839	0.025			0.7	0.993	0.991	0.012			0.7	1.000	1.000	0.002
		0.8	0.942	0.905	0.022			0.8	0.999	0.996	0.007			0.8	1.000	1.000	0.001
0.9	0.964	0.948	0.020	0.9	1.000	1.000	0.003	0.9	1.000	1.000	0.001						
50	50	0	0.074	0.046	0.050	100	50	0	0.053	0.043	0.050	200	50	0	0.065	0.047	0.050
		0.1	0.061	0.067	0.048			0.1	0.078	0.071	0.044			0.1	0.100	0.086	0.047
		0.2	0.067	0.093	0.044			0.2	0.199	0.174	0.046			0.2	0.383	0.306	0.045
		0.3	0.127	0.157	0.041			0.3	0.434	0.353	0.044			0.3	0.765	0.662	0.038
		0.4	0.268	0.276	0.041			0.4	0.721	0.598	0.040			0.4	0.938	0.876	0.032
		0.5	0.457	0.417	0.042			0.5	0.873	0.781	0.040			0.5	0.993	0.969	0.027
		0.6	0.638	0.567	0.040			0.6	0.946	0.895	0.037			0.6	0.999	0.995	0.020
		0.7	0.780	0.680	0.039			0.7	0.977	0.948	0.032			0.7	1.000	1.000	0.014
		0.8	0.872	0.782	0.033			0.8	0.993	0.971	0.027			0.8	1.000	1.000	0.012
0.9	0.938	0.856	0.029	0.9	0.999	0.989	0.024	0.9	1.000	1.000	0.010						
50	100	0	0.072	0.049	0.050	100	100	0	0.050	0.041	0.050	200	100	0	0.059	0.043	0.050
		0.1	0.075	0.051	0.051			0.1	0.056	0.046	0.056			0.1	0.084	0.071	0.050
		0.2	0.080	0.073	0.054			0.2	0.095	0.109	0.060			0.2	0.295	0.219	0.054
		0.3	0.126	0.116	0.055			0.3	0.279	0.240	0.064			0.3	0.680	0.501	0.053
		0.4	0.228	0.203	0.053			0.4	0.580	0.434	0.061			0.4	0.914	0.779	0.051
		0.5	0.405	0.310	0.050			0.5	0.799	0.641	0.062			0.5	0.986	0.921	0.047
		0.6	0.609	0.438	0.049			0.6	0.917	0.792	0.056			0.6	1.000	0.971	0.036
		0.7	0.757	0.567	0.049			0.7	0.971	0.885	0.050			0.7	1.000	0.997	0.033
		0.8	0.863	0.691	0.047			0.8	0.989	0.942	0.048			0.8	1.000	0.999	0.026
0.9	0.925	0.794	0.046	0.9	0.994	0.980	0.044	0.9	1.000	1.000	0.021						
50	200	0	0.069	0.050	0.050	100	200	0	0.056	0.047	0.050	200	200	0	0.057	0.042	0.050
		0.1	0.065	0.053	0.051			0.1	0.058	0.053	0.053			0.1	0.066	0.069	0.056
		0.2	0.073	0.071	0.052			0.2	0.072	0.085	0.050			0.2	0.160	0.159	0.053
		0.3	0.105	0.094	0.049			0.3	0.224	0.167	0.051			0.3	0.529	0.354	0.057
		0.4	0.179	0.152	0.051			0.4	0.457	0.290	0.050			0.4	0.844	0.600	0.054
		0.5	0.334	0.213	0.052			0.5	0.711	0.452	0.048			0.5	0.956	0.825	0.048
		0.6	0.519	0.305	0.052			0.6	0.887	0.620	0.049			0.6	0.990	0.934	0.047
		0.7	0.680	0.413	0.051			0.7	0.951	0.753	0.048			0.7	0.999	0.977	0.046
		0.8	0.806	0.520	0.052			0.8	0.980	0.853	0.046			0.8	1.000	0.988	0.043
0.9	0.888	0.632	0.051	0.9	0.994	0.918	0.046	0.9	1.000	1.000	0.040						
50	500	0	0.052	0.050	0.050	100	500	0	0.052	0.048	0.050	200	500	0	0.053	0.039	0.050
		0.1	0.050	0.054	0.053			0.1	0.048	0.050	0.045			0.1	0.054	0.048	0.051
		0.2	0.054	0.058	0.051			0.2	0.050	0.073	0.044			0.2	0.109	0.099	0.048
		0.3	0.081	0.070	0.049			0.3	0.152	0.123	0.041			0.3	0.446	0.233	0.048
		0.4	0.155	0.099	0.052			0.4	0.415	0.186	0.040			0.4	0.794	0.440	0.045
		0.5	0.304	0.146	0.052			0.5	0.670	0.300	0.041			0.5	0.943	0.628	0.044
		0.6	0.467	0.196	0.056			0.6	0.850	0.461	0.042			0.6	0.988	0.787	0.042
		0.7	0.613	0.263	0.057			0.7	0.940	0.591	0.043			0.7	0.997	0.895	0.042
		0.8	0.751	0.358	0.055			0.8	0.979	0.716	0.044			0.8	1.000	0.960	0.041
0.9	0.841	0.448	0.058	0.9	0.994	0.798	0.043	0.9	1.000	0.985	0.039						

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Table S.3. Empirical power in situations of different sparsity levels (signal number ranging from 0 to 9) under the Gaussian error setting. Tests are conducted at the nominal level $\alpha = 0.05$.

T	N	β	N_0	PE-PGCT	DH	HPI	T	N	β	N_0	PE-PGCT	DH	HPI	T	N	β	N_0	PE-PGCT	DH	HPI
50	5	0.5	0	0.06	0.05	0.05	100	5	0.4	0	0.044	0.05	0.05	200	5	0.3	0	0.061	0.05	0.05
			1	0.813	0.795	0.015					0.928	0.932	0.002					0.959	0.954	0.005
			2	0.986	0.984	0.046					1	1	0.027					0.998	0.998	0.032
			3	0.998	0.998	0.201					1	1	0.196					1	1	0.244
			4	1	1	0.593					1	1	0.651					1	1	0.712
			5	1	1	1					1	1	1					1	1	1
50	10	0.5	0	0.052	0.05	0.05	100	10	0.4	0	0.041	0.049	0.05	200	10	0.3	0	0.052	0.049	0.05
			1	0.742	0.714	0.025					0.873	0.87	0.013					0.904	0.895	0.024
			2	0.966	0.962	0.056					0.994	0.994	0.025					0.998	0.998	0.051
			3	0.995	0.994	0.125					0.999	0.999	0.084					1	1	0.154
			4	1	1	0.294					1	1	0.226					1	1	0.404
			5	1	1	0.564					1	1	0.52					1	1	0.732
			6	1	1	0.763					1	1	0.754					1	1	0.89
			7	1	1	0.906					1	1	0.923					1	1	0.975
			8	1	1	0.995					1	1	1					1	1	1
			9	1	1	0.999					1	1	1					1	1	1
50	20	0.5	0	0.062	0.049	0.05	100	20	0.4	0	0.056	0.047	0.05	200	20	0.3	0	0.06	0.047	0.05
			1	0.666	0.606	0.036					0.793	0.754	0.032					0.86	0.831	0.031
			2	0.934	0.922	0.053					0.976	0.974	0.058					0.993	0.992	0.06
			3	0.985	0.984	0.103					0.999	0.999	0.128					1	1	0.15
			4	0.997	0.996	0.179					1	1	0.226					1	1	0.291
			5	1	1	0.311					1	1	0.414					1	1	0.511
			6	1	1	0.478					1	1	0.611					1	1	0.71
			7	1	1	0.67					1	1	0.79					1	1	0.89
			8	1	1	0.869					1	1	0.945					1	1	0.986
			9	1	1	0.935					1	1	0.988					1	1	0.997
50	50	0.5	0	0.074	0.046	0.05	100	50	0.4	0	0.053	0.043	0.05	200	50	0.3	0	0.065	0.047	0.05
			1	0.457	0.417	0.042					0.721	0.598	0.04					0.765	0.662	0.038
			2	0.684	0.746	0.058					0.926	0.897	0.064					0.953	0.94	0.068
			3	0.807	0.893	0.095					0.98	0.977	0.128					0.998	0.997	0.119
			4	0.878	0.974	0.15					0.998	0.998	0.195					1	1	0.208
			5	0.927	0.994	0.249					1	1	0.304					1	1	0.33
			6	0.957	0.999	0.334					1	1	0.431					1	1	0.466
			7	0.972	1	0.456					1	1	0.584					1	1	0.606
			8	0.99	1	0.61					1	1	0.747					1	1	0.799
			9	0.996	1	0.725					1	1	0.852					1	1	0.881
50	100	0.5	0	0.072	0.049	0.05	100	100	0.4	0	0.05	0.041	0.05	200	100	0.3	0	0.059	0.043	0.05
			1	0.405	0.31	0.05					0.58	0.434	0.061					0.68	0.501	0.053
			2	0.604	0.603	0.065					0.77	0.766	0.091					0.91	0.851	0.079
			3	0.699	0.794	0.081					0.855	0.914	0.135					0.977	0.961	0.124
			4	0.775	0.916	0.115					0.903	0.978	0.19					0.996	0.993	0.19
			5	0.845	0.974	0.16					0.948	0.997	0.258					1	1	0.296
			6	0.878	0.989	0.201					0.963	0.999	0.325					1	1	0.378
			7	0.92	0.997	0.252					0.978	1	0.422					1	1	0.503
			8	0.961	1	0.347					0.988	1	0.55					1	1	0.653
			9	0.975	1	0.434					0.993	1	0.653					1	1	0.756
50	200	0.5	0	0.069	0.05	0.05	100	200	0.4	0	0.056	0.047	0.05	200	200	0.3	0	0.057	0.042	0.05
			1	0.334	0.213	0.052					0.457	0.29	0.05					0.529	0.354	0.057
			2	0.492	0.433	0.063					0.685	0.596	0.063					0.741	0.708	0.078
			3	0.6	0.631	0.074					0.777	0.801	0.086					0.823	0.882	0.105
			4	0.706	0.793	0.098					0.85	0.919	0.112					0.886	0.975	0.136
			5	0.789	0.901	0.134					0.907	0.971	0.157					0.925	0.991	0.187
			6	0.841	0.95	0.162					0.941	0.99	0.209					0.953	0.999	0.261
			7	0.881	0.984	0.201					0.959	0.999	0.284					0.979	1	0.33
			8	0.924	0.996	0.249					0.982	1	0.368					0.991	1	0.432
			9	0.94	0.997	0.292					0.984	1	0.447					0.995	1	0.519
50	500	0.5	0	0.052	0.05	0.05	100	500	0.4	0	0.052	0.048	0.05	200	500	0.3	0	0.053	0.039	0.05
			1	0.304	0.146	0.052					0.415	0.186	0.04					0.446	0.233	0.048
			2	0.461	0.262	0.064					0.596	0.413	0.055					0.654	0.484	0.049
			3	0.54	0.398	0.076					0.683	0.567	0.069					0.727	0.662	0.061
			4	0.613	0.537	0.085					0.752	0.738	0.086					0.805	0.839	0.084
			5	0.679	0.695	0.098					0.842	0.864	0.123					0.886	0.93	0.123
			6	0.729	0.793	0.117					0.874	0.93	0.146					0.916	0.974	0.149
			7	0.795	0.878	0.139					0.922	0.971	0.175					0.947	0.991	0.186
			8	0.865	0.95	0.174					0.964	0.992	0.226					0.978	0.999	0.25
			9	0.885	0.967	0.194					0.969	0.998	0.261					0.984	1	0.285

Power-Boosted Granger-Causal Discovery for Large Heterogeneous Panel Data

Table S.4. Empirical power under sparse alternatives ($N_0 = 1$) under the non-Gaussian setting. Tests are conducted at level $\alpha = 0.05$.

T	N	β	PE-PGCT	DH	HPJ	T	N	β	PE-PGCT	DH	HPJ	T	N	β	PE-PGCT	DH	HPJ
50	5	0	0.059	0.050	0.050	100	5	0	0.046	0.050	0.050	200	5	0	0.058	0.050	0.050
		0.1	0.109	0.094	0.054			0.1	0.170	0.194	0.057			0.1	0.288	0.277	0.038
		0.2	0.286	0.258	0.045			0.2	0.526	0.558	0.041			0.2	0.775	0.765	0.017
		0.3	0.531	0.498	0.026			0.3	0.808	0.825	0.019			0.3	0.962	0.959	0.003
		0.4	0.735	0.699	0.016			0.4	0.941	0.947	0.007			0.4	0.992	0.992	0.002
		0.5	0.869	0.852	0.003			0.5	0.975	0.976	0.001			0.5	0.998	0.998	0.001
		0.6	0.925	0.914	0.002			0.6	0.991	0.992	0.001			0.6	0.999	0.999	0.000
		0.7	0.956	0.948	0.002			0.7	0.999	0.999	0.001			0.7	0.999	0.999	0.000
		0.8	0.979	0.973	0.001			0.8	0.999	0.999	0.001			0.8	0.999	0.999	0.000
0.9	0.989	0.988	0.001	0.9	1.000	1.000	0.000	0.9	1.000	1.000	0.000						
50	10	0	0.047	0.050	0.050	100	10	0	0.051	0.050	0.050	200	10	0	0.049	0.049	0.050
		0.1	0.091	0.091	0.054			0.1	0.128	0.126	0.056			0.1	0.202	0.212	0.049
		0.2	0.221	0.208	0.060			0.2	0.399	0.380	0.043			0.2	0.692	0.694	0.025
		0.3	0.414	0.403	0.046			0.3	0.691	0.679	0.027			0.3	0.914	0.915	0.014
		0.4	0.628	0.612	0.036			0.4	0.880	0.864	0.020			0.4	0.982	0.981	0.004
		0.5	0.796	0.772	0.023			0.5	0.954	0.948	0.012			0.5	0.995	0.995	0.000
		0.6	0.880	0.871	0.014			0.6	0.981	0.979	0.007			0.6	0.999	0.999	0.000
		0.7	0.928	0.920	0.009			0.7	0.990	0.990	0.003			0.7	0.999	0.999	0.000
		0.8	0.953	0.945	0.007			0.8	0.995	0.995	0.003			0.8	1.000	1.000	0.000
0.9	0.973	0.968	0.005	0.9	0.997	0.997	0.001	0.9	1.000	1.000	0.000						
50	20	0	0.070	0.050	0.050	100	20	0	0.051	0.044	0.050	200	20	0	0.038	0.043	0.050
		0.1	0.085	0.073	0.049			0.1	0.105	0.102	0.048			0.1	0.151	0.161	0.062
		0.2	0.165	0.132	0.037			0.2	0.313	0.283	0.045			0.2	0.540	0.547	0.056
		0.3	0.358	0.281	0.036			0.3	0.629	0.578	0.037			0.3	0.861	0.854	0.039
		0.4	0.565	0.485	0.030			0.4	0.824	0.790	0.033			0.4	0.963	0.960	0.026
		0.5	0.722	0.651	0.021			0.5	0.930	0.900	0.026			0.5	0.992	0.991	0.018
		0.6	0.838	0.767	0.018			0.6	0.969	0.962	0.018			0.6	0.996	0.995	0.009
		0.7	0.910	0.870	0.016			0.7	0.985	0.983	0.017			0.7	0.999	0.999	0.002
		0.8	0.946	0.916	0.012			0.8	0.989	0.989	0.016			0.8	0.999	1.000	0.001
0.9	0.963	0.949	0.012	0.9	0.994	0.991	0.013	0.9	1.000	1.000	0.000						
50	50	0	0.071	0.048	0.050	100	50	0	0.074	0.044	0.050	200	50	0	0.045	0.039	0.050
		0.1	0.066	0.060	0.042			0.1	0.102	0.076	0.053			0.1	0.103	0.104	0.052
		0.2	0.084	0.104	0.043			0.2	0.246	0.182	0.054			0.2	0.427	0.384	0.048
		0.3	0.190	0.185	0.044			0.3	0.521	0.399	0.048			0.3	0.794	0.708	0.037
		0.4	0.377	0.316	0.038			0.4	0.757	0.625	0.049			0.4	0.942	0.910	0.032
		0.5	0.543	0.466	0.036			0.5	0.885	0.789	0.043			0.5	0.986	0.969	0.029
		0.6	0.717	0.619	0.033			0.6	0.947	0.889	0.040			0.6	0.992	0.989	0.027
		0.7	0.837	0.742	0.031			0.7	0.971	0.942	0.032			0.7	0.996	0.994	0.018
		0.8	0.905	0.823	0.031			0.8	0.990	0.966	0.026			0.8	0.997	0.996	0.014
0.9	0.949	0.887	0.030	0.9	0.995	0.983	0.023	0.9	0.999	0.997	0.012						
50	100	0	0.065	0.048	0.050	100	100	0	0.061	0.045	0.050	200	100	0	0.055	0.038	0.050
		0.1	0.063	0.058	0.049			0.1	0.061	0.059	0.049			0.1	0.083	0.065	0.053
		0.2	0.082	0.082	0.051			0.2	0.128	0.119	0.046			0.2	0.321	0.225	0.047
		0.3	0.158	0.150	0.050			0.3	0.368	0.255	0.046			0.3	0.718	0.549	0.044
		0.4	0.313	0.248	0.052			0.4	0.617	0.476	0.049			0.4	0.913	0.785	0.047
		0.5	0.532	0.376	0.051			0.5	0.815	0.686	0.046			0.5	0.969	0.916	0.045
		0.6	0.674	0.512	0.051			0.6	0.910	0.828	0.042			0.6	0.990	0.967	0.040
		0.7	0.802	0.657	0.051			0.7	0.950	0.901	0.039			0.7	0.995	0.986	0.033
		0.8	0.863	0.752	0.048			0.8	0.973	0.945	0.037			0.8	1.000	0.994	0.031
0.9	0.912	0.805	0.046	0.9	0.985	0.959	0.034	0.9	1.000	0.998	0.026						
50	200	0	0.056	0.049	0.050	100	200	0	0.057	0.048	0.050	200	200	0	0.052	0.041	0.050
		0.1	0.054	0.049	0.050			0.1	0.059	0.064	0.050			0.1	0.055	0.058	0.048
		0.2	0.070	0.070	0.052			0.2	0.086	0.101	0.045			0.2	0.212	0.149	0.049
		0.3	0.123	0.111	0.050			0.3	0.293	0.187	0.042			0.3	0.610	0.384	0.046
		0.4	0.252	0.165	0.048			0.4	0.566	0.350	0.042			0.4	0.859	0.641	0.040
		0.5	0.430	0.264	0.048			0.5	0.755	0.525	0.042			0.5	0.948	0.817	0.040
		0.6	0.604	0.384	0.048			0.6	0.876	0.676	0.041			0.6	0.984	0.926	0.040
		0.7	0.748	0.497	0.047			0.7	0.945	0.798	0.041			0.7	0.993	0.969	0.037
		0.8	0.828	0.608	0.047			0.8	0.974	0.873	0.040			0.8	0.997	0.987	0.036
0.9	0.887	0.705	0.048	0.9	0.987	0.927	0.040	0.9	0.998	0.989	0.035						
50	500	0	0.059	0.050	0.050	100	500	0	0.057	0.049	0.050	200	500	0	0.052	0.041	0.050
		0.1	0.059	0.053	0.049			0.1	0.058	0.059	0.051			0.1	0.053	0.050	0.048
		0.2	0.064	0.065	0.048			0.2	0.080	0.090	0.051			0.2	0.153	0.106	0.046
		0.3	0.122	0.089	0.047			0.3	0.239	0.140	0.049			0.3	0.514	0.232	0.049
		0.4	0.237	0.129	0.047			0.4	0.515	0.250	0.049			0.4	0.805	0.451	0.044
		0.5	0.407	0.198	0.045			0.5	0.733	0.405	0.050			0.5	0.925	0.653	0.044
		0.6	0.563	0.279	0.045			0.6	0.855	0.543	0.051			0.6	0.969	0.803	0.042
		0.7	0.700	0.365	0.046			0.7	0.918	0.679	0.052			0.7	0.985	0.908	0.043
		0.8	0.809	0.458	0.047			0.8	0.958	0.786	0.051			0.8	0.994	0.943	0.043
0.9	0.871	0.542	0.047	0.9	0.977	0.860	0.049	0.9	0.996	0.968	0.041						

Power-Boosted Granger-Causal Discovery for Large Heterogeneous Panel Data

Table S.5. Empirical power in situations of different sparsity levels (signal number ranging from 0 to 9) under the non-Gaussian setting. Tests are conducted at the nominal level $\alpha = 0.05$.

T	N	β	N_0	PE-PGCT	DH	HPI	T	N	β	N_0	PE-PGCT	DH	HPI	T	N	β	N_0	PE-PGCT	DH	HPI
50	5	0.5	0	0.059	0.050	0.050	100	5	0.4	0	0.046	0.050	0.050	200	5	0.3	0	0.069	0.063	0.056
			1	0.869	0.852	0.003				1	0.941	0.947	0.007				1	0.931	0.930	0.005
			2	0.987	0.985	0.044				2	0.998	0.998	0.056				2	0.998	0.998	0.076
			3	0.999	0.999	0.225				3	1.000	1.000	0.288				3	1.000	1.000	0.342
			4	1.000	1.000	0.662				4	1.000	1.000	0.724				4	1.000	1.000	0.753
			5	1.000	1.000	0.992				5	1.000	1.000	1.000				5	1.000	1.000	0.999
50	10	0.5	0	0.047	0.050	0.050	100	10	0.4	0	0.051	0.050	0.050	200	10	0.3	0	0.049	0.049	0.050
			1	0.796	0.772	0.023				1	0.880	0.864	0.020				1	0.914	0.915	0.014
			2	0.975	0.974	0.053				2	0.993	0.993	0.051				2	0.997	0.997	0.041
			3	0.994	0.994	0.123				3	1.000	1.000	0.147				3	0.999	0.999	0.113
			4	1.000	1.000	0.288				4	1.000	1.000	0.350				4	1.000	1.000	0.312
			5	1.000	1.000	0.572				5	1.000	1.000	0.684				5	1.000	1.000	0.667
			6	1.000	1.000	0.763				6	1.000	1.000	0.863				6	1.000	1.000	0.859
			7	1.000	1.000	0.904				7	1.000	1.000	0.962				7	1.000	1.000	0.968
			8	1.000	1.000	0.993				8	1.000	1.000	0.999				8	1.000	1.000	1.000
9	1.000	1.000	0.997	9	1.000	1.000	1.000	9	1.000	1.000	1.000									
50	20	0.5	0	0.070	0.050	0.050	100	20	0.4	0	0.051	0.044	0.050	200	20	0.3	0	0.038	0.043	0.050
			1	0.722	0.651	0.021				1	0.824	0.790	0.033				1	0.861	0.854	0.039
			2	0.950	0.932	0.047				2	0.982	0.977	0.061				2	0.986	0.986	0.075
			3	0.991	0.987	0.084				3	0.999	0.999	0.117				3	0.998	0.998	0.171
			4	0.995	0.994	0.173				4	1.000	1.000	0.240				4	1.000	1.000	0.334
			5	1.000	1.000	0.300				5	1.000	1.000	0.445				5	1.000	1.000	0.591
			6	1.000	1.000	0.457				6	1.000	1.000	0.648				6	1.000	1.000	0.785
			7	1.000	1.000	0.659				7	1.000	1.000	0.849				7	1.000	1.000	0.929
			8	1.000	1.000	0.859				8	1.000	1.000	0.972				8	1.000	1.000	0.994
9	1.000	1.000	0.944	9	1.000	1.000	0.992	9	1.000	1.000	1.000									
50	50	0.5	0	0.071	0.048	0.050	100	50	0.4	0	0.074	0.044	0.050	200	50	0.3	0	0.045	0.039	0.050
			1	0.543	0.466	0.036				1	0.757	0.625	0.049				1	0.794	0.708	0.037
			2	0.790	0.826	0.057				2	0.945	0.917	0.084				2	0.966	0.964	0.068
			3	0.881	0.944	0.087				3	0.995	0.991	0.145				3	0.996	0.996	0.124
			4	0.938	0.993	0.146				4	1.000	1.000	0.226				4	1.000	1.000	0.221
			5	0.969	0.998	0.241				5	1.000	1.000	0.350				5	1.000	1.000	0.363
			6	0.982	1.000	0.340				6	1.000	1.000	0.481				6	1.000	1.000	0.504
			7	0.993	1.000	0.423				7	1.000	1.000	0.618				7	1.000	1.000	0.653
			8	0.998	1.000	0.591				8	1.000	1.000	0.797				8	1.000	1.000	0.815
9	0.999	1.000	0.699	9	1.000	1.000	0.882	9	1.000	1.000	0.902									
50	100	0.5	0	0.065	0.048	0.050	100	100	0.4	0	0.061	0.045	0.050	200	100	0.3	0	0.055	0.038	0.050
			1	0.532	0.376	0.051				1	0.617	0.476	0.049				1	0.718	0.549	0.044
			2	0.739	0.705	0.077				2	0.833	0.844	0.068				2	0.923	0.875	0.073
			3	0.827	0.875	0.101				3	0.913	0.952	0.099				3	0.978	0.970	0.116
			4	0.895	0.949	0.138				4	0.945	0.992	0.144				4	0.997	0.994	0.182
			5	0.944	0.987	0.189				5	0.976	0.998	0.211				5	1.000	0.999	0.269
			6	0.963	0.996	0.245				6	0.988	1.000	0.284				6	1.000	1.000	0.356
			7	0.981	0.999	0.318				7	0.991	1.000	0.366				7	1.000	1.000	0.454
			8	0.992	1.000	0.434				8	0.997	1.000	0.506				8	1.000	1.000	0.638
9	0.996	1.000	0.504	9	0.999	1.000	0.598	9	1.000	1.000	0.743									
50	200	0.5	0	0.056	0.049	0.050	100	200	0.4	0	0.057	0.048	0.050	200	200	0.3	0	0.052	0.041	0.050
			1	0.430	0.264	0.048				1	0.566	0.350	0.042				1	0.610	0.384	0.046
			2	0.630	0.536	0.056				2	0.772	0.692	0.050				2	0.803	0.730	0.064
			3	0.746	0.754	0.078				3	0.859	0.869	0.069				3	0.879	0.899	0.092
			4	0.820	0.878	0.091				4	0.921	0.959	0.095				4	0.925	0.974	0.123
			5	0.898	0.949	0.116				5	0.958	0.991	0.137				5	0.967	0.993	0.172
			6	0.930	0.975	0.156				6	0.967	0.996	0.198				6	0.975	1.000	0.227
			7	0.955	0.989	0.215				7	0.983	0.999	0.271				7	0.987	1.000	0.306
			8	0.977	0.999	0.292				8	0.995	1.000	0.355				8	0.996	1.000	0.429
9	0.983	0.999	0.340	9	0.998	1.000	0.419	9	0.998	1.000	0.528									
50	500	0.5	0	0.059	0.050	0.050	100	500	0.4	0	0.057	0.049	0.050	200	500	0.3	0	0.052	0.041	0.050
			1	0.407	0.198	0.045				1	0.515	0.250	0.049				1	0.514	0.232	0.049
			2	0.598	0.373	0.052				2	0.728	0.521	0.052				2	0.725	0.503	0.052
			3	0.703	0.526	0.059				3	0.820	0.719	0.058				3	0.814	0.703	0.066
			4	0.787	0.675	0.067				4	0.877	0.847	0.072				4	0.871	0.856	0.086
			5	0.853	0.797	0.078				5	0.885	0.886	0.096				5	0.932	0.950	0.114
			6	0.893	0.872	0.091				6	0.919	0.945	0.113				6	0.949	0.977	0.140
			7	0.925	0.937	0.111				7	0.952	0.981	0.135				7	0.971	0.994	0.180
			8	0.958	0.978	0.145				8	0.977	0.991	0.173				8	0.986	0.999	0.225
9	0.966	0.988	0.170	9	0.980	0.996	0.206	9	0.988	1.000	0.272									

C.2. Computational Costs

To provide an empirical assessment of the computational cost, we report in Table S.6 the median implementation time per replication, computed over 100 runs. As shown in the table, when $N < T - K$, PE-PGCT adds negligible computational cost relative to the pivotal test DH, since computing J_{PE} requires little additional effort once the individual statistics W_i have been obtained for DH. When $N \geq T - K$, PE-PGCT incurs extra time to compute J_{PE} ; however, the added complexity scales linearly with N .

Table S.6. Median time (in seconds) per replication (over 100 runs).

N	$T = 50$				$T = 100$				$T = 200$			
	DH	HPJ	J_{PE}	PE-PGCT	DH	HPJ	J_{PE}	PE-PGCT	DH	HPJ	J_{PE}	PE-PGCT
5	0.009	0.002	1.3×10^{-5}	0.009	0.010	0.003	1.3×10^{-5}	0.010	0.011	0.005	1.4×10^{-5}	0.011
10	0.019	0.003	1.4×10^{-5}	0.019	0.018	0.005	1.4×10^{-5}	0.018	0.021	0.011	1.5×10^{-5}	0.021
20	0.035	0.007	1.4×10^{-5}	0.035	0.038	0.012	1.5×10^{-5}	0.038	0.044	0.027	1.6×10^{-5}	0.044
50	0.079	0.021	0.078	0.099	0.088	0.035	1.6×10^{-5}	0.088	0.102	0.072	1.6×10^{-5}	0.102
100	0.160	0.043	0.158	0.201	0.175	0.079	0.172	0.252	0.203	0.170	1.7×10^{-5}	0.203
200	0.316	0.111	0.313	0.424	0.346	0.214	0.344	0.558	0.405	0.420	0.403	0.824
500	0.785	0.436	0.785	1.221	0.864	0.818	0.868	1.687	1.040	1.747	1.039	2.786

Note: J_{PE} reports the computation time for the PE component only, whereas DH, HPJ, and PE-PGCT report the total runtime of their respective tests.

C.3. Comparisons with Additional Relevant Approaches

We consider comparisons with two additional benchmarks: (i) a max-form test, whose asymptotic distribution for panel Granger-causality testing follows from Proposition 3.5 as detailed in Remark 3.6, and (ii) the method of Fan et al. (2015).

C.3.1. A STANDALONE MAX TEST BASED ON M_{Wald}

Proposition 3.5 can also be used to derive the asymptotic distribution of the max-form statistic $M_{Wald} := \max_{1 \leq i \leq N} W_i$, which may itself serve as a standalone testing procedure. Under the same conditions of Proposition 3.5 and suppose $\log N = o(T^{1/7})$ with small and finite K , we have, under the null hypothesis H_0 , $P(M_{Wald} - 2 \log N - (K-2) \log \log N \leq x) \rightarrow \exp(-\exp(-x/2))$ as $N, T \rightarrow \infty$.

C.3.2. THE POWER ENHANCEMENT METHOD OF FAN ET AL. (2015)

Since Fan et al. (2015) considered a different testing problem, we first adapt it to the panel Granger causality framework to enable direct comparisons. Following the key insights of the paper, a power enhancement component can be constructed as the sum of marginal statistics whose absolute values exceed a given threshold. Here,

$$\tilde{J}_{PE} = \sqrt{NK} \sum_{i=1}^N \sum_{j=1}^K \frac{|\hat{\beta}_{ij}|}{\sqrt{\widehat{var}(\hat{\beta}_{ij})}} I \left\{ \frac{|\hat{\beta}_{ij}|}{\sqrt{\widehat{var}(\hat{\beta}_{ij})}} > \tilde{\delta}_{N,T} \right\}.$$

There are a few challenges in determining an appropriate value $\tilde{\delta}_{N,T}$ in this setting. First, by design, $\tilde{\delta}_{N,T}$ must dominate the maximum noise level $\max_{1 \leq i \leq N} \max_{1 \leq j \leq K} |\hat{\beta}_{ij}| / \sqrt{\widehat{var}(\hat{\beta}_{ij})}$ under H_0 . Even under the assumption of cross-sectional independence across i , accounting for the dependence across lags j remains difficult, complicating the determination of a theoretically justified threshold. Second, extending Fan et al. (2015), we expect $\tilde{\delta}_{N,T}$ to take the form of

$$\tilde{\delta}_{N,T} = \zeta \cdot \sqrt{\log(NK)} \cdot \log \log T \quad (\text{S.49})$$

for some constant $\zeta > 0$. Simulations show that the empirical performance of this approach is highly sensitive to the value of ζ , making it challenging to ensure reliable finite-sample performance.

C.3.3. EMPIRICAL COMPARISONS

We adopt the same data-generating process as in Section 4. Table S.7 summarizes the comparison among PE-PGCT, MAX, and Fan2015 in terms of their empirical Type I error rates. The results show that the MAX test maintains proper type I error rate control when T is moderate or large and N is large. When T is small but N is large, the test exhibits a slight inflation in Type I error. We also observe that it tends to be conservative when N is small.

For power assessment, we again consider two scenarios: (i) a sparse alternative with $N_0 = 1$ and $\beta = 0.5$, and (ii) a relatively denser alternative with $N_0 = 5$ and $\beta = 0.3$. The comparisons of empirical power are reported in Table S.8. The MAX test performs particularly well under sparse alternatives, but is less effective under dense alternatives especially when N is small. This aligns with the intuition that the maximum statistic is sensitive to strong marginal signals, making it well-suited for detecting sparse signals. However, it may lose power in dense settings where signals are weak but spread across many coordinates. Comparing PE-PGCT with Fan2015, PE-PGCT slightly underperforms Fan2015 when T is small, but outperforms it when T is moderate or large, particularly in settings with a large N . Overall, the two deliver comparable performance. Methodologically, the proposed PE-PGCT offers several advantages in addressing the aforementioned challenges of threshold determination. It not only provides theoretically rigorous guarantees but also delivers tuning-free, practical recommendations for strong finite-sample performance.

 Table S.7. Additional comparisons of empirical Type I error under the Gaussian error setting (nominal level $\alpha = 0.05$).

T	N	PE-PGCT	MAX	$\tilde{J}_{PE}$	T	N	PE-PGCT	MAX	$\tilde{J}_{PE}$	T	N	PE-PGCT	MAX	$\tilde{J}_{PE}$
50	5	0.060	0.026	0.085	100	5	0.044	0.019	0.048	200	5	0.061	0.025	0.061
	10	0.052	0.035	0.075		10	0.041	0.035	0.042		10	0.052	0.024	0.052
	20	0.062	0.049	0.069		20	0.056	0.043	0.056		20	0.060	0.039	0.060
	50	0.074	0.063	0.082		50	0.053	0.039	0.053		50	0.065	0.037	0.065
	100	0.072	0.081	0.077		100	0.050	0.038	0.050		100	0.059	0.030	0.059
	200	0.069	0.083	0.073		200	0.056	0.059	0.055		200	0.057	0.038	0.057
	500	0.052	0.129	0.055		500	0.052	0.064	0.052		500	0.053	0.033	0.053

Note: For the implementation of $\tilde{J}_{PE}$, we set $\zeta = 1.5$ in (S.49).

 Table S.8. Additional comparisons of empirical power under the Gaussian error setting (nominal level $\alpha = 0.05$).

(i) (a sparse alternative) $N_0 = 1, \beta = 0.5$														
T	N	PE-PGCT	MAX	$\tilde{J}_{PE}$	T	N	PE-PGCT	MAX	$\tilde{J}_{PE}$	T	N	PE-PGCT	MAX	$\tilde{J}_{PE}$
50	5	0.813	0.791	0.843	100	5	0.985	0.980	0.989	200	5	0.999	0.999	0.999
	10	0.742	0.764	0.779		10	0.958	0.965	0.964		10	0.999	0.999	0.999
	20	0.666	0.727	0.715		20	0.925	0.962	0.925		20	0.998	1	0.997
	50	0.457	0.671	0.547		50	0.873	0.940	0.862		50	0.993	0.999	0.984
	100	0.405	0.680	0.476		100	0.799	0.928	0.755		100	0.986	1	0.967
	200	0.334	0.604	0.384		200	0.711	0.916	0.636		200	0.956	0.995	0.915
	500	0.304	0.590	0.331		500	0.670	0.906	0.579		500	0.943	0.991	0.879
(ii) (a dense alternative) $N_0 = 5, \beta = 0.3$														
T	N	PE-PGCT	MAX	$\tilde{J}_{PE}$	T	N	PE-PGCT	MAX	$\tilde{J}_{PE}$	T	N	PE-PGCT	MAX	$\tilde{J}_{PE}$
50	5	0.988	0.849	0.935	100	5	1	0.994	1	200	5	1	1	1
	10	0.959	0.800	0.926		10	1	0.989	1		10	1	1	1
	20	0.910	0.755	0.899		20	0.998	0.981	0.998		20	1	1	1
	50	0.401	0.648	0.463		50	0.985	0.948	0.985		50	0.998	0.997	0.997
	100	0.294	0.564	0.345		100	0.669	0.927	0.601		100	1	1	1
	200	0.218	0.507	0.238		200	0.554	0.891	0.424		200	0.925	1	0.831
	500	0.147	0.455	0.161		500	0.418	0.851	0.284		500	0.886	0.999	0.674

Note: Given the practical challenge of selecting an appropriate threshold value in (S.49), we determine the cutoff for $\tilde{J}_{PE}$ empirically and report its size-adjusted power to ensure a fair comparison.

D. Supplemental Data Documentation for Section 5

Table S.9 reports the summary statistics for the 220 small banks and 80 large banks. Table S.10 provides detailed information about the CERTs and bank names of the $N = 300$ financial institutions used in the real data analysis in Section 5.

Table S.9. Summary Statistics of Bank Size by Group

Group	N	Mean	SD	Min	Mdn	Max
Small	220	10.455	0.701	8.726	10.477	11.630
Large	80	12.833	1.400	11.654	12.466	18.305
All	300	11.089	1.410	8.726	10.888	18.305

Table S.10. CERT Identifiers and Bank Names for the $N = 300$ Banks

CERT	NAME	CERT	NAME	CERT	NAME
58	UNITED BANK	77	BANKSOUTH	89	SIGNATURE BANK OF ARKANSAS
99	BANK OF DELIGHT	197	ALLIANCE BANK	275	MORGANTOWN BANK&TRUST CO INC
659	FIVE STAR BANK	803	WESBANCO BANK INC	899	GERBER STATE BANK
976	BLISSFIELD STATE BANK	1049	FIRST NB OF STEELEVILLE	1117	AMERICAN BANK&TRUST
1208	GUARANTY BANK&TRUST N A	1219	WELLINGTON STATE BANK	1391	ARLINGTON STATE BANK
1413	PERENNIAL BANK	1468	BANK OF CROCKETT	1536	BANK OF STOCKTON
1545	FIRST NB AMES IOWA	1562	PEOPLES BANK	1622	FIRST SECURITY BANK
1686	FARMERS STATE BANK	1745	FARMERS&MERCHANTS BANK	1833	STATE BANK
1865	KANSAS STB OVERBROOK KANSAS	1900	FIRST INDEPENDENT BANK	2160	FIRST NB OF WAYNESBORO
2301	FIRST STATE BANK	2315	ARMSTRONG BANK	2448	PIONEER COMMUNITY BANK INC
2608	NATIONAL GRAND BK MARBLEHEAD	2682	FIRST NB OF BROOKSVILLE	2754	FARMERS NB OF LEBANON
2767	FIRST NB OF RUSSELL SPRINGS	2784	FIRST BANK OF BOAZ	3034	FIRST NATIONAL BANK COLORADO
3099	FIRST NB OF BELLVILLE	3183	NEWFIRST NATIONAL BANK	3232	GRANGER NATIONAL BANK
3276	AUSTIN BANK TEXAS NA	3293	FIRST TEXAS BANK	3301	FIRST LIBERTY BANK
3376	OZONA BANK	3657	FIRST AMERICAN BANK	3664	INB NATIONAL ASSN
3747	FIRST NEIGHBOR BANK NA	3887	FIRST NATIONAL BANK	4222	FIRST FARMERS NB OF WAURIKA
4255	CAMDEN NATIONAL BANK	4297	CAPITAL ONE NATIONAL ASSN	4444	FIRST NB IN CRESTON
4477	FIRST NB OF MANNING	4506	SECURITY NB OF SIOUX CITY IA	4636	FIRST HERITAGE BANK
4770	ELEVATE BANK NATIONAL ASSN	4832	SHORE UNITED BANK NA	4865	SANDY SPRING BANK
4958	FIRST NB OF ONEIDA	5058	SUPERIOR NATIONAL BANK	5197	COMMUNITY FIRST BANK
5198	FIRST NB OF MILACA	5280	WNB FINANCIAL N A	5432	FIRST NORTHEAST BANK OF NE
5434	MNB BANK	5538	FIRST NB OF STERLING CITY	5589	FANNIN BANK
5628	PIGGOTT STATE BANK	5632	LOGAN COUNTY BANK	5633	FIRST SECURITY BANK
5686	BANK OF EDISON	5716	FARMERS&MERCHANTS BANK	5730	STATE BANK OF ST JACOB
5892	KALAMAZOO COUNTY STATE BANK	5956	PEOPLES BANK	6180	BANK OF MONROE
6194	CITIZENS STATE BANK	6354	FIRST HOPE BK A NATL BK ASSN	6384	PNC BANK NATIONAL ASSN
6548	U S BANK NATIONAL ASSN	6584	COMMUNITY FIRST BANK NA	6623	LCNB NATIONAL BANK
6821	NATIONAL BANK OF BLACKSBURG	6917	FIRST BANK	6954	CAYUGA LAKE NATIONAL BANK
6959	BALLSTON SPA NATIONAL BANK	7085	FIRST NB OF GROTON	7799	CITIZENS&NORTHERN BANK
7888	FIRST NB OF PENNSYLVANIA	7910	MARS BANK	8214	FARMERS STATE BANK OF HAMEL
8276	ALTON BANK	8317	FIRST STATE BANK OF FORSYTH	8499	UNION STATE BANK
8594	MONTROSE SAVINGS BANK	8745	FIRST STB OF MIDDLEBURY	8811	HEBRON SAVINGS BANK
8840	PEOPLES STATE BANK OF WELLS	8866	MERCHANTS BANK NATIONAL ASSN	8904	NODAWAY VALLEY BANK
8980	SECURITY STB OF OKLAHOMA	9099	SOUTH OTTUMWA SAVINGS BANK	9244	GLENNVILLE BANK
9322	CROSS KEYS BANK	9340	FARMERS&MERCHANTS STB NY MIL	9385	BANK OF BROOKFIELD PURDIN NA
9620	FIRST BANK	9632	FORTRESS BANK	9871	ST HENRY BANK
9889	MID PENN BANK	9932	COMMERCIAL STB OF WAGNER	9978	FIRST STATE BANK OF UVALDE
9998	CAPON VALLEY BANK	10011	WOODFORD STATE BANK	10012	PORTAGE COUNTY BANK
10159	FIRST STATE BANK&TRUST	10185	ELYSIAN BANK	10248	FIRST INTERNATIONAL B&T
10315	BANK OF HALLS	10363	CITIZENS&FARMERS BANK	10438	RARITAN STATE BANK
10492	FARMERS&MERCHANTS SB	10533	WINNSBORO STATE B&T CO	10595	COMMUNITY FIRST BANK
10672	EXCHANGE BANK	10709	CROSSROADS BANK	10719	PEOPLES STB OF HALLETTSVILLE
10788	ILLINI STATE BANK	10831	FIRST T&SB OF WATSEKA IL	10886	LEGENCE BANK
10973	ALLIANCE BANK	11002	ST CLAIR COUNTY STATE BANK	11063	FIRST-CITIZENS BANK&TRUST CO
11124	S&T BANK	11352	SANBORN SAVINGS BANK	11358	CUMBERLAND SECURITY BANK INC
11446	UNITY BANK OF MISSISSIPPI	11475	THAYER COUNTY BANK	11477	AUBURN STATE BANK
11492	INBANK	11527	FIRST STB OF POND CREEK OK	11580	BANK OF SOUTHSIDE VIRGINIA

Table S.10. CERT Identifiers and Bank Names for the $N = 300$ Banks (continued)

CERT	NAME	CERT	NAME	CERT	NAME
11680	FARMERS STB OF WESTERN IL	11686	STATE BANK OF CHERRY	11777	COMMERCIAL BANK
12283	FIRST STATE BANK	12309	CITIZENS BANK NATIONAL ASSN	12372	FOWLER STATE BANK
12384	JERSEY STATE BANK	12395	LENA STATE BANK	12410	LA SALLE STATE BANK
12422	SECURITY STATE BANK	12444	STATE BANK OF COLD SPRING	12511	ALLIANCE BANK
12531	PEOPLES BANK	12591	CONVERSE COUNTY BANK	12594	PUEBLO BANK&TRUST CO
12614	ORIGIN BANK	12758	FARMERS STB ALLEN OKLAHOMA	12858	FARMERS TRUST&SAVINGS BANK
13109	JACKSON COUNTY BANK	13112	OWEN COUNTY STATE BANK	13113	PEOPLES TRUST&SAVINGS BANK
13264	BANK OF LUXEMBURG	13297	SOLVAY BANK	13339	STATE BANK&TRUST CO
13397	BANK OF UTICA	13413	KAW VALLEY STATE BANK	13662	FARMERS&MERCHANTS STB BLOOMF
13774	BENTON COUNTY STATE BANK	13813	FIRST NB IN PINCKNEYVILLE	13879	PEOPLESBANK CODORUS VALLEY
13883	FARMERS STATE BANK	13951	COMMUNITY BANK OF LOUISIANA	14017	MIDDLETOWN VALLEY BANK
14036	MISSOURI BANK	14079	GNB BANK	14269	CITIZENS STATE BANK
14359	JERSEY SHORE STATE BANK	14392	FARMERS TRUST&SAVINGS BANK	14507	CNB BANK INC
14581	CLARKSON BANK	14660	BANK OF PRAIRIE DU SAC	14679	INDUSTRIAL BANK
14692	PEOPLES BANK	14695	MA BANK	14772	DANVILLE STATE SAVINGS BANK
14843	MIDWESTONE BANK	14853	FIRST NB AT ST JAMES	14952	LYON COUNTY STATE BANK
15010	COLUMBUS BANK&TRUST CO	15092	RICHLAND COUNTY BANK	15136	BANK OF WISCONSIN DELLS
15271	FIRST STATE BANK	15478	UNITED VALLEY BANK	15590	TAYLOR COUNTY BANK
15593	IPSWICH STATE BANK	15651	STATE BANK&TRUST CO	15771	NORTHEAST GEORGIA BANK
15908	UNION STATE BANK	15932	CITIZENS BANK	15945	FOREST PARK NB&T CO
15995	PCSB BANK	16039	SAWYER SAVINGS BANK	16075	RAYNE STATE BANK&TRUST CO
16161	CAMPBELL COUNTY BANK INC	16177	FIRST STATE BANK	16231	DAKOTA HERITAGE BANK
16413	BANK OF STRONGHURST	16600	FIRST STATE BANK	16723	CAROLINA BANK&TRUST CO
16840	RIVER FALLS STATE BANK	16895	FIRST STATE BANK OF BEDIAS	16908	SPIRO STATE BANK
17145	HERITAGE BANK	17230	ALBANY B&T CO NA	17350	FIRST STB OF SAN DIEGO
17491	PLAINSCAPITAL BANK	17513	FISHER NATIONAL BANK	17534	KEYBANK NATIONAL ASSN
17741	MAINE COMMUNITY BANK	17749	BATH SAVINGS INSTITUTION	17769	BANK OF HAYS
17811	FIRST NB IN PORT LAVACA	17832	KAW VALLEY STATE B&T CO	17906	AMERICAN BANK&TRUST CO
17960	MERRIMACK COUNTY SB	18019	PATRIOTS BANK	18132	PEARLAND STATE BANK
18198	ION BANK	18258	THOMASTON SAVINGS BANK	18384	NORTHEAST SECURITY BANK
18568	COMMUNITY BANK OF EASTON	18677	VICTORY BANK	18922	TRI CITY NATIONAL BANK
18930	BANK OF BELLE GLADE	18983	RIVER CITY BANK	19101	HARFORD BANK
19184	HILLTOP NATIONAL BANK	19220	ANB BANK	19230	COMMUNITY NB IN MONMOUTH
19333	REPUBLIC BANK OF CHICAGO	19396	UNISON BANK	19459	UNITED COMMUNITY BANK
19603	HOMETOWN BANK NATIONAL ASSN	19689	STATE BANK OF WHITTINGTON	19904	CENTINEL BANK OF TAOS
20040	MERCHANTS&MANUFACTURERS BANK	20215	BCBANK INC	20241	AMERICAN BANK NATIONAL ASSN
20268	SCALE BANK	20369	HEARTLAND BANK&TRUST CO	21083	UNITED MISSISSIPPI BANK
21379	FIRST NATURALSTATE BANK	21752	FIRST STATE BANK	21848	QUAIL CREEK BANK
22090	COFFEE COUNTY BANK	22551	FIRST TEXAS BANK	22597	BANK OF THE SIERRA
22621	TOWN COUNTRY UNITED BANK	23007	OREGON PACIFIC BK DBA OR PAC	23286	FALL RIVER FIVE CENTS SB
23293	FLORENCE BANK	23620	DEDHAM INST FOR SVG	23764	FMS BANK
23772	COMMERCE BANK	24922	GRAND VALLEY BANK	25161	COMMUNITY NB OF OKARCHE
25249	C3BANK NATIONAL ASSN	25291	BANK OF THE VALLEY	25665	COMMUNITY NATIONAL BANK
25883	FIRST EAGLE BANK	26181	TEXAS NB OF JACKSONVILLE	26333	GENESEE REGIONAL BANK
26451	FIDELITY COOP BANK	26590	ABINGTON BANK	26602	CHARLES RIVER BANK
26620	READING COOP BANK	26647	FIRSTTRUST SAVINGS BANK	27847	PIONEER SAVINGS BANK
28222	HADDON SAVINGS BANK	28729	ASCENDIA BANK	28747	HOYNE SAVINGS BANK
28925	SECURITY SAVINGS BANK	29009	LIBERTY BANK FOR SAVINGS	29093	INVESTORS COMMUNITY BANK
29809	FIRST SB OF HEGEWISCH	30516	CROSSBRIDGE COMMUNITY BANK	30585	MAYVILLE SAVINGS BANK
30619	ABBEVILLE FIRST BANK SSB	31936	FAYETTE SAVINGS BANK SSB	32135	SHELBY SAVINGS BANK SSB
32838	COMMUNITY FIRST BANK INC	33440	CITIZENS FIRST BANK	33503	UNITY BANK
34040	FIRST AMERICAN STATE BANK	34207	GREAT PLAINS NATIONAL BANK	34363	CORE BANK
34422	AMERICAN BANK	34666	COMMUNITY BANK	34685	PEOPLES BANK OF COMMERCE